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Du et al.

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(45) **Date of Patent: Sep. 2, 2025**

(54) **REKEYING END-TO-END EFFICIENT ENCRYPTION WITH SECURITY CHAINING**

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(73) Assignee: **Dell Products, L.P.**, Round Rock, TX (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 380 days.

(21) Appl. No.: **17/816,217**

(22) Filed: **Jul. 29, 2022**

(65) **Prior Publication Data**

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(51) **Int. Cl.**
H04L 9/08 (2006.01)
H04L 9/14 (2006.01)

(52) **U.S. Cl.**
CPC ***H04L 9/0891*** (2013.01); ***H04L 9/14*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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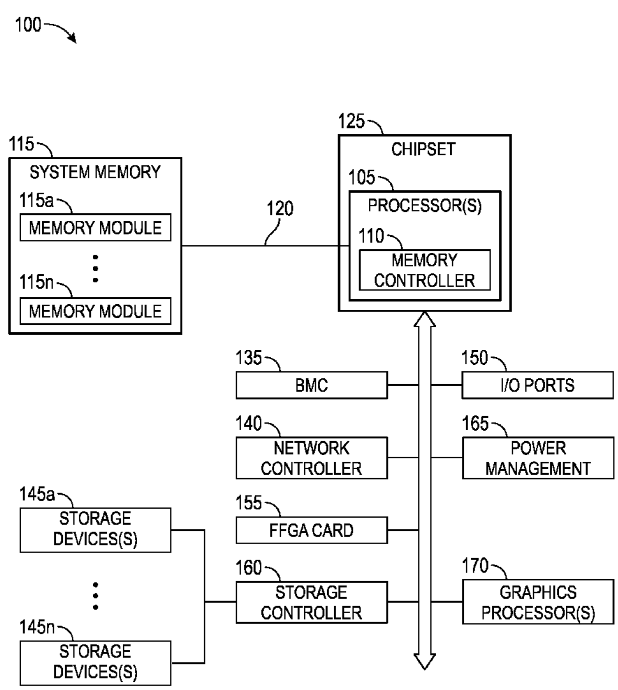
(57) **ABSTRACT**

Rekeying an Information Handling System (IHS) network End-to-End Efficient Encryption (E2EEE) with security chaining includes locking an encrypted data volume, preventing reading of, and writing to, the encrypted data volume by applications. A data source IHS may request a new key from a key management system and write new metadata in a trailer of the encrypted data block using a different key slot than a currently used and active metadata key slot, wherein the different key slot is updated with the with the new key. The data source IHS then sends an out-of-data signal to change the use key slot from the currently used key slot to the different key slot. Thereafter, the data source IHS unlocks the encrypted data volume, enabling writing and/or reading user data by the data source IHS and encryption and decryption in all IHS E2EEE data connection segment interfaces.

20 Claims, 23 Drawing Sheets



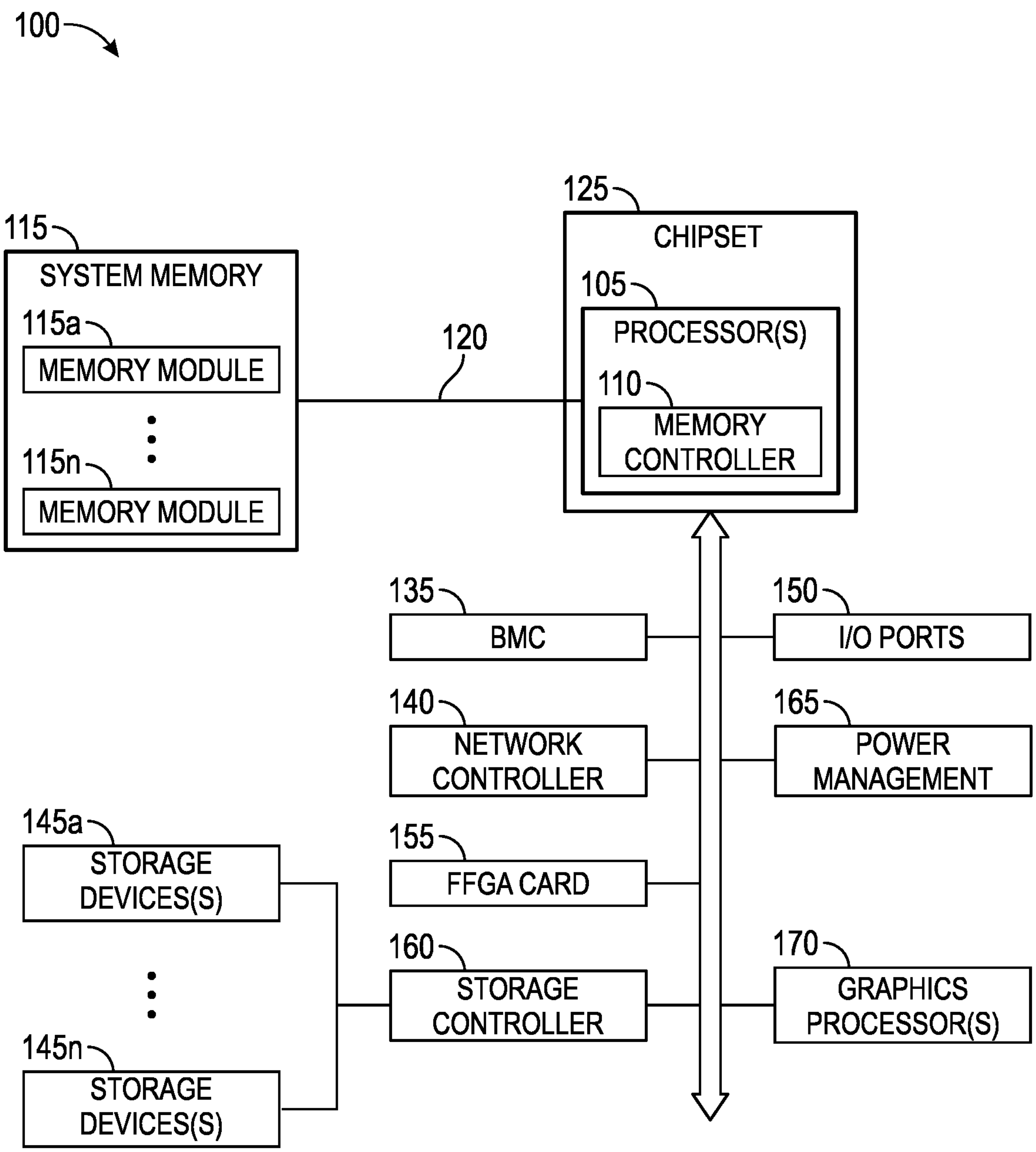

100
115
SYSTEM MEMORY
115a
MEMORY MODULE
115n
MEMORY MODULE
120
125
CHIPSET
105
PROCESSOR(S)
110
MEMORY CONTROLLER
135
BMC
150
I/O PORTS
140
NETWORK CONTROLLER
165
POWER MANAGEMENT
145a
STORAGE DEVICES(S)
155
FFGA CARD
160
STORAGE CONTROLLER
170
GRAPHICS PROCESSOR(S)
145n
STORAGE DEVICES(S)

FIG. 1

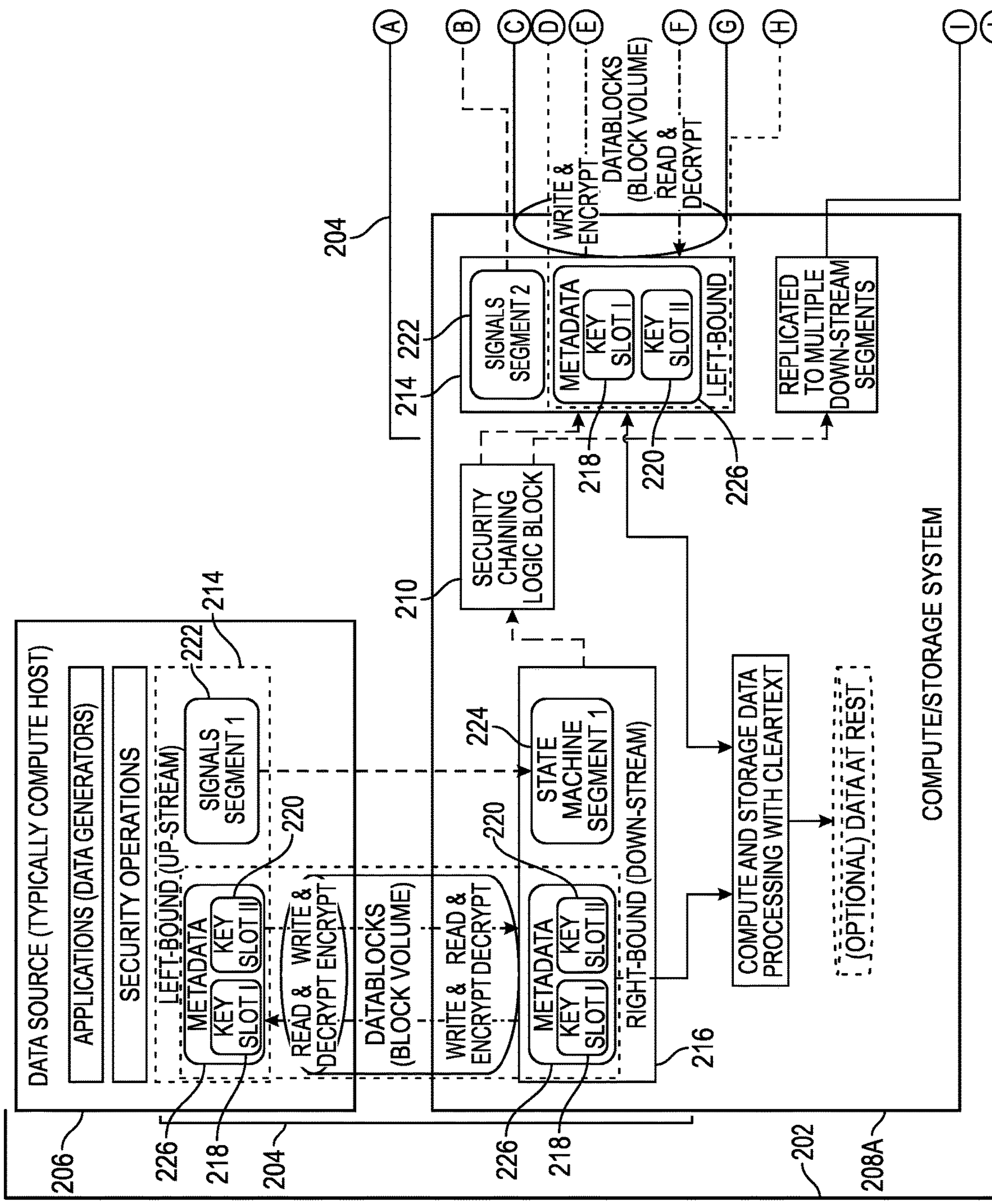
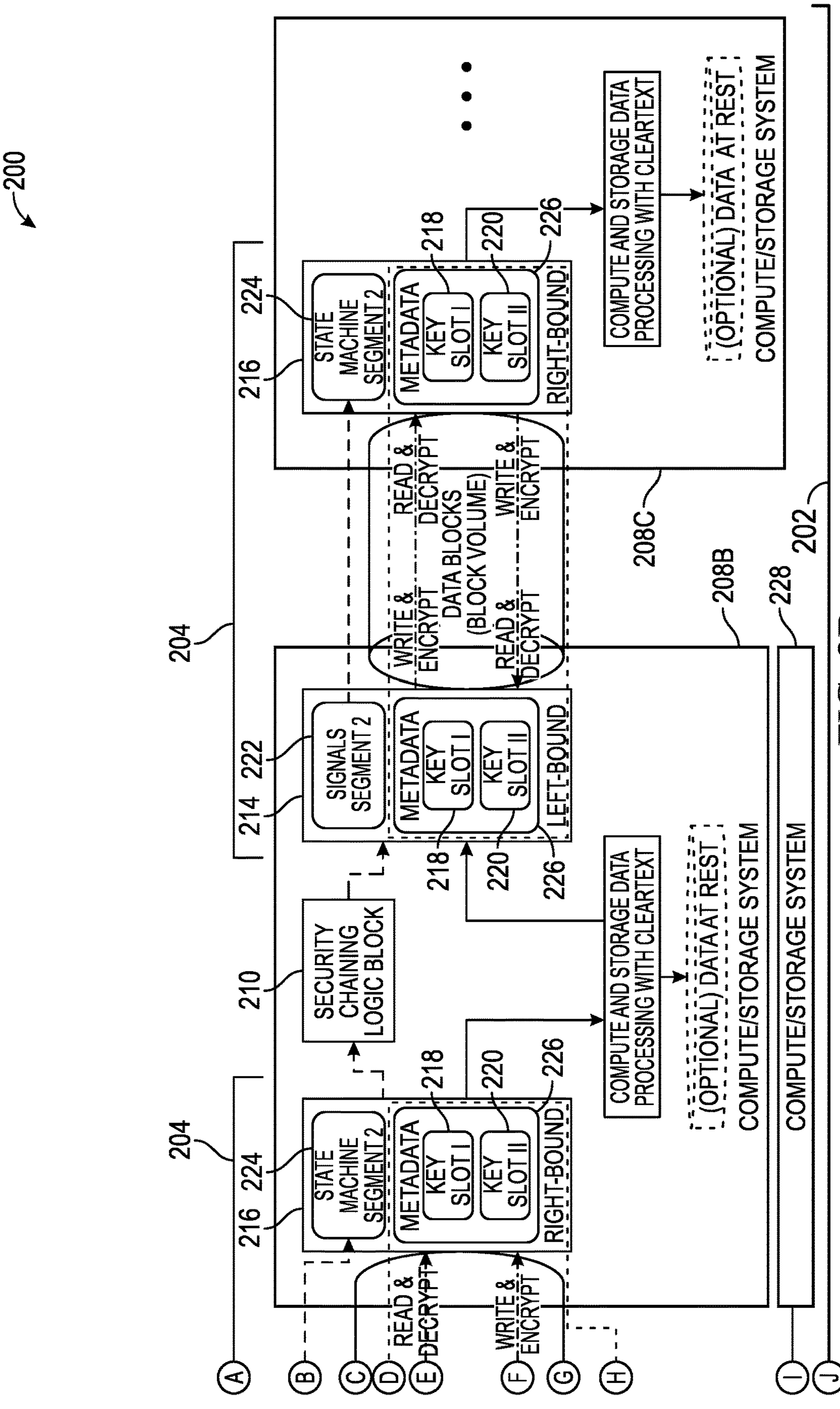

DATA SOURCE (TYPICALLY COMPUTE HOST)
APPLICATIONS (DATA GENERATORS)
SECURITY OPERATIONS
LEFT-BOUND (UP-STREAM)
SIGNALS SEGMENT 1
METADATA
KEY SLOT I
KEY SLOT II
READ & DECRYPT
WRITE & ENCRYPT
DATABLOCKS (BLOCK VOLUME)
WRITE & ENCRYPT
READ & DECRYPT
METADATA
KEY SLOT I
KEY SLOT II
RIGHT-BOUND (DOWN-STREAM)
STATE MACHINE SEGMENT 1
SECURITY CHAINING LOGIC BLOCK
COMPUTE AND STORAGE DATA PROCESSING WITH CLEARTEXT
(OPTIONAL) DATA AT REST
COMPUTE/STORAGE SYSTEM
SIGNALS SEGMENT 2
METADATA
KEY SLOT I
KEY SLOT II
LEFT-BOUND
WRITE & ENCRYPT
DATABLOCKS (BLOCK VOLUME)
READ & DECRYPT
REPLICATED TO MULTIPLE DOWN-STREAM SEGMENTS
A
B
C
D
E
F
G
H
I
J
202
204
206
208A
210
214
216
218
220
222
224
226

FIG. 2A



200
204
216
224
STATE MACHINE SEGMENT 2
METADATA
KEY SLOT I
KEY SLOT II
RIGHT-BOUND
218
220
226
210
SECURITY CHAINING LOGIC BLOCK
214
222
SIGNALS SEGMENT 2
LEFT-BOUND
COMPUTE AND STORAGE DATA PROCESSING WITH CLEARTEXT
(OPTIONAL) DATA AT REST
COMPUTE/STORAGE SYSTEM
WRITE & ENCRYPT
READ & DECRYPT
DATA BLOCKS (BLOCK VOLUME)
208B
208C
228
202
A
B
C
D
E
F
G
H
I
J

FIG. 2B

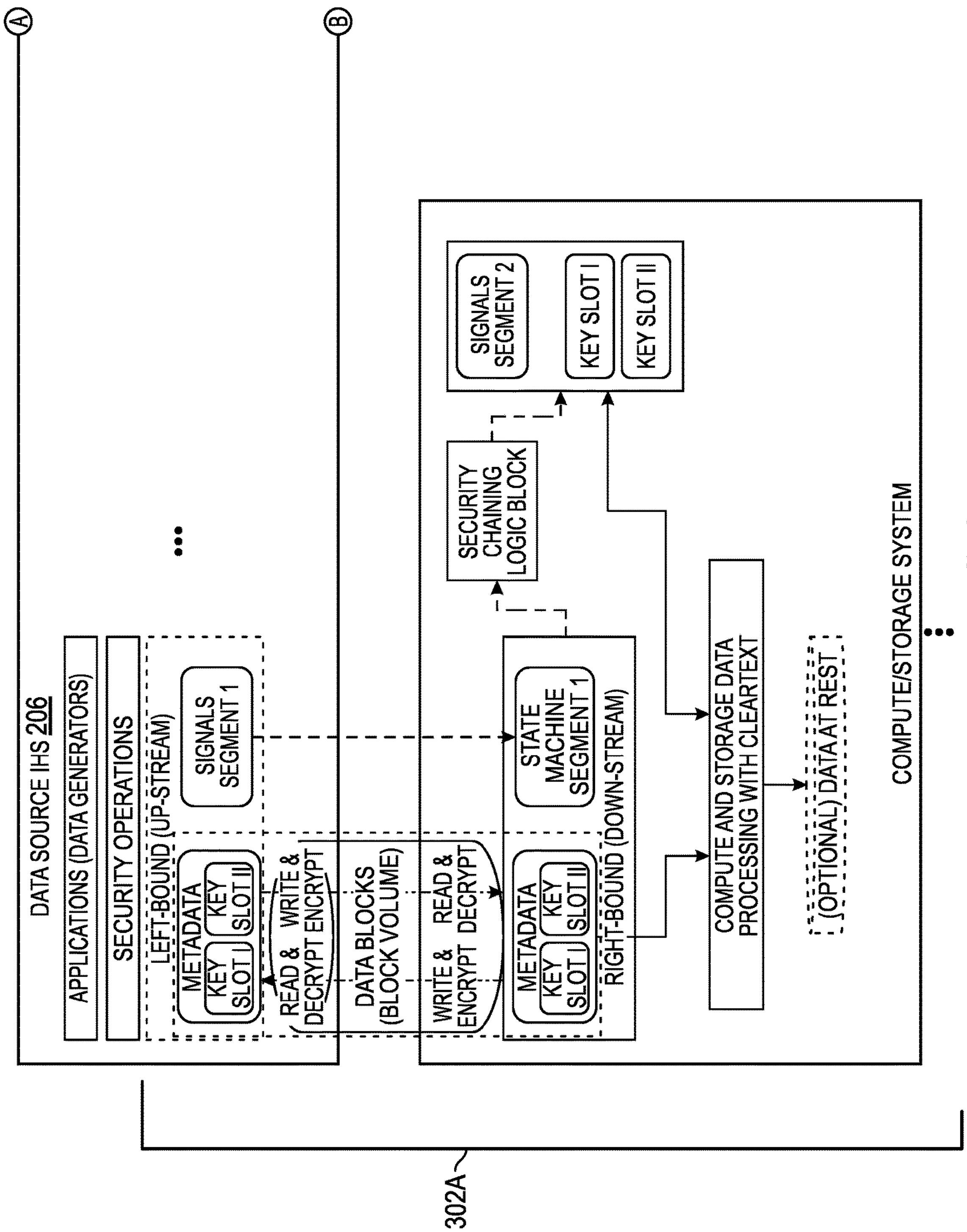

A
B
DATA SOURCE IHS 206
APPLICATIONS (DATA GENERATORS)
SECURITY OPERATIONS
LEFT-BOUND (UP-STREAM)
METADATA
KEY SLOT I
KEY SLOT II
SIGNALS SEGMENT 1
READ & DECRYPT
WRITE & ENCRYPT
DATA BLOCKS (BLOCK VOLUME)
WRITE & ENCRYPT
READ & DECRYPT
METADATA
KEY SLOT I
KEY SLOT II
RIGHT-BOUND (DOWN-STREAM)
STATE MACHINE SEGMENT 1
SECURITY CHAINING LOGIC BLOCK
SIGNALS SEGMENT 2
KEY SLOT I
KEY SLOT II
COMPUTE AND STORAGE DATA PROCESSING WITH CLEARTEXT
(OPTIONAL) DATA AT REST
COMPUTE/STORAGE SYSTEM
302A

FIG. 3A

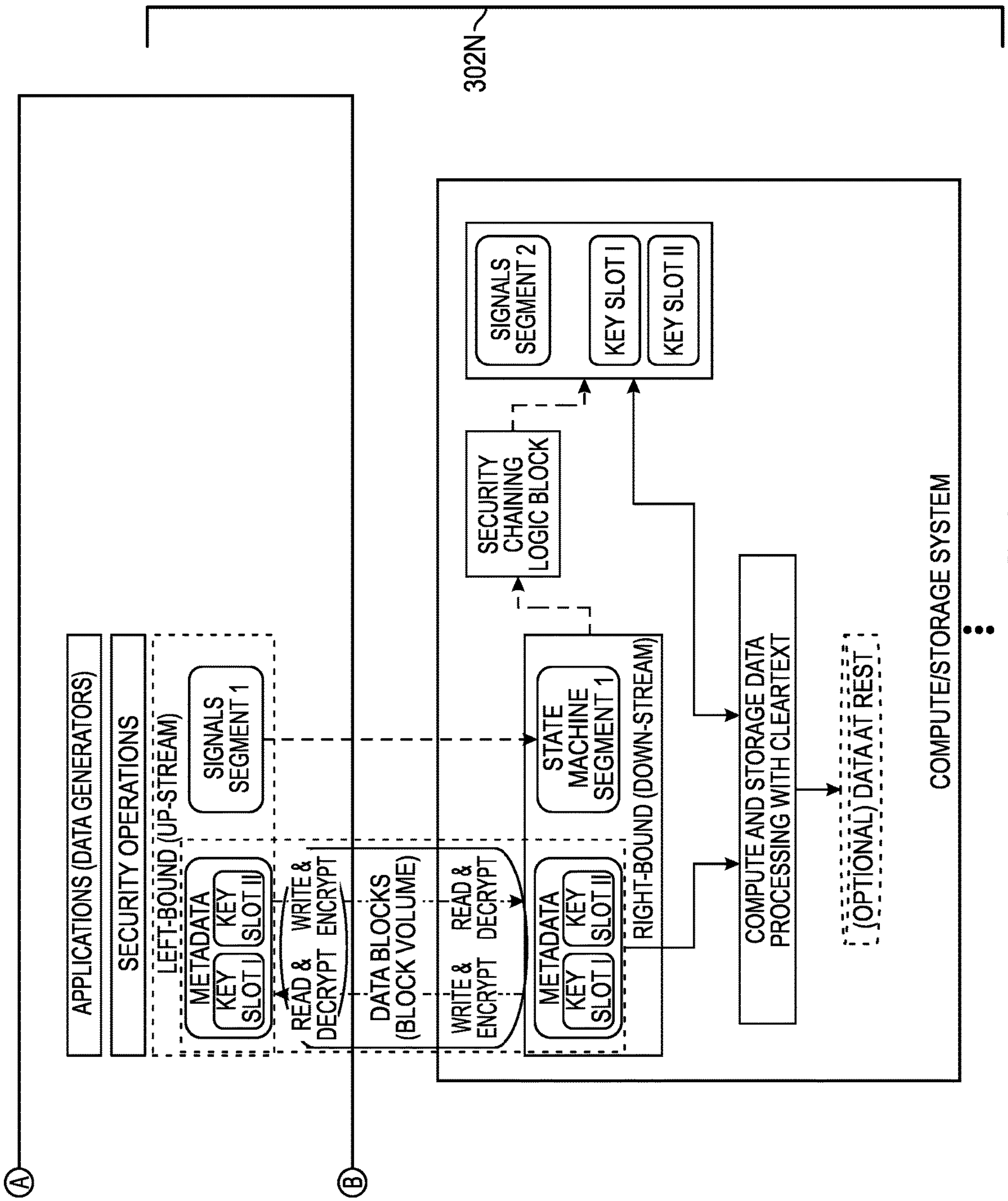

302N
APPLICATIONS (DATA GENERATORS)
SECURITY OPERATIONS
LEFT-BOUND (UP-STREAM)
METADATA
KEY SLOT I
KEY SLOT II
SIGNALS SEGMENT 1
READ & DECRYPT
WRITE & ENCRYPT
DATA BLOCKS (BLOCK VOLUME)
WRITE & ENCRYPT
READ & DECRYPT
METADATA
KEY SLOT I
KEY SLOT II
STATE MACHINE SEGMENT 1
RIGHT-BOUND (DOWN-STREAM)
SECURITY CHAINING LOGIC BLOCK
SIGNALS SEGMENT 2
KEY SLOT I
KEY SLOT II
COMPUTE AND STORAGE DATA PROCESSING WITH CLEARTEXT
(OPTIONAL) DATA AT REST
COMPUTE/STORAGE SYSTEM
A
B

FIG. 3B

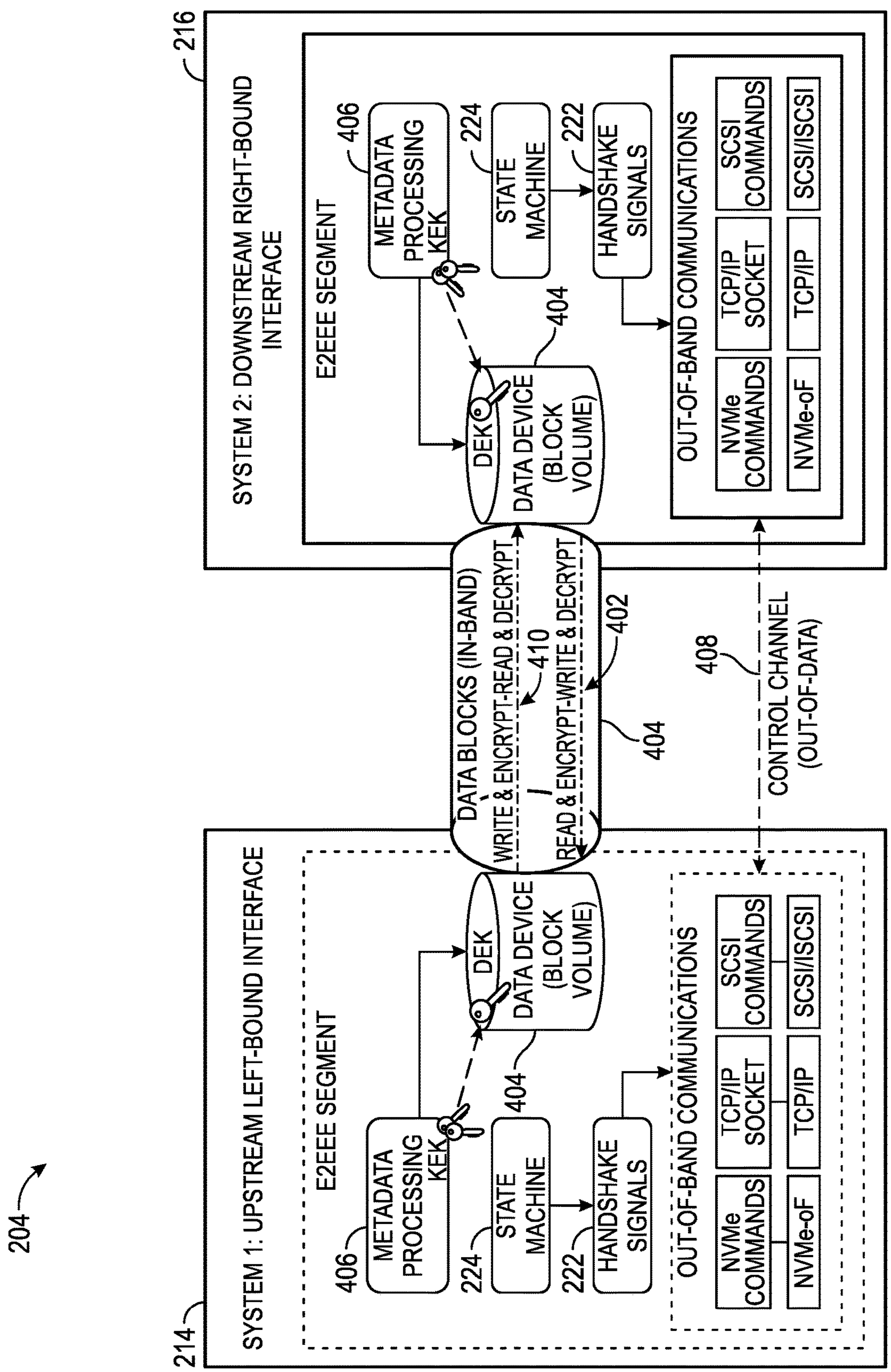

204
214
SYSTEM 1: UPSTREAM LEFT-BOUND INTERFACE
E2EEE SEGMENT
406
METADATA PROCESSING KEK
224
STATE MACHINE
222
HANDSHAKE SIGNALS
404
DEK
DATA DEVICE (BLOCK VOLUME)
OUT-OF-BAND COMMUNICATIONS
NVMe COMMANDS
TCP/IP SOCKET
SCSI COMMANDS
NVMe-oF
TCP/IP
SCSI/ISCSI
DATA BLOCKS (IN-BAND)
WRITE & ENCRYPT-READ & DECRYPT
410
READ & ENCRYPT-WRITE & DECRYPT
402
404
408
CONTROL CHANNEL (OUT-OF-DATA)
216
SYSTEM 2: DOWNSTREAM RIGHT-BOUND INTERFACE
E2EEE SEGMENT
406
METADATA PROCESSING KEK
224
STATE MACHINE
222
HANDSHAKE SIGNALS
404
DEK
DATA DEVICE (BLOCK VOLUME)
OUT-OF-BAND COMMUNICATIONS
NVMe COMMANDS
TCP/IP SOCKET
SCSI COMMANDS
NVMe-oF
TCP/IP
SCSI/ISCSI

FIG. 4

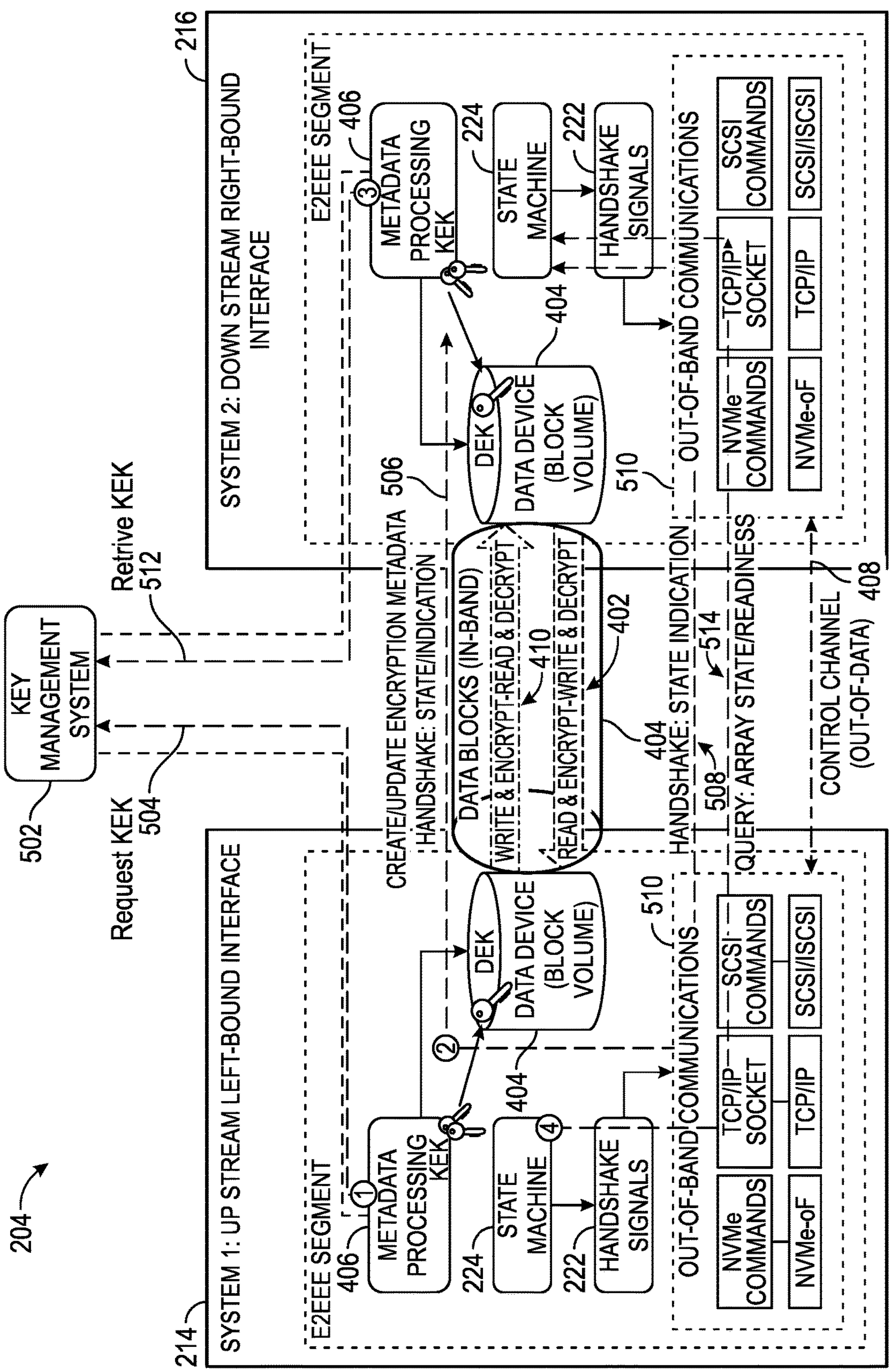

204
KEY MANAGEMENT SYSTEM
502
Request KEK
504
Retrive KEK
512
SYSTEM 1: UP STREAM LEFT-BOUND INTERFACE
214
E2EEE SEGMENT
406
METADATA PROCESSING KEK
224
STATE MACHINE
222
HANDSHAKE SIGNALS
404
DEK
DATA DEVICE (BLOCK VOLUME)
510
OUT-OF-BAND COMMUNICATIONS
NVMe COMMANDS
TCP/IP SOCKET
SCSI COMMANDS
NVMe-oF
TCP/IP
SCSI/ISCSI
SYSTEM 2: DOWN STREAM RIGHT-BOUND INTERFACE
216
CREATE/UPDATE ENCRYPTION METADATA
HANDSHAKE: STATE/INDICATION
506
DATA BLOCKS (IN-BAND)
WRITE & ENCRYPT-READ & DECRYPT
410
READ & ENCRYPT-WRITE & DECRYPT
402
HANDSHAKE: STATE INDICATION
508
514
QUERY: ARRAY STATE/READINESS
CONTROL CHANNEL (OUT-OF-DATA)
408

FIG. 5

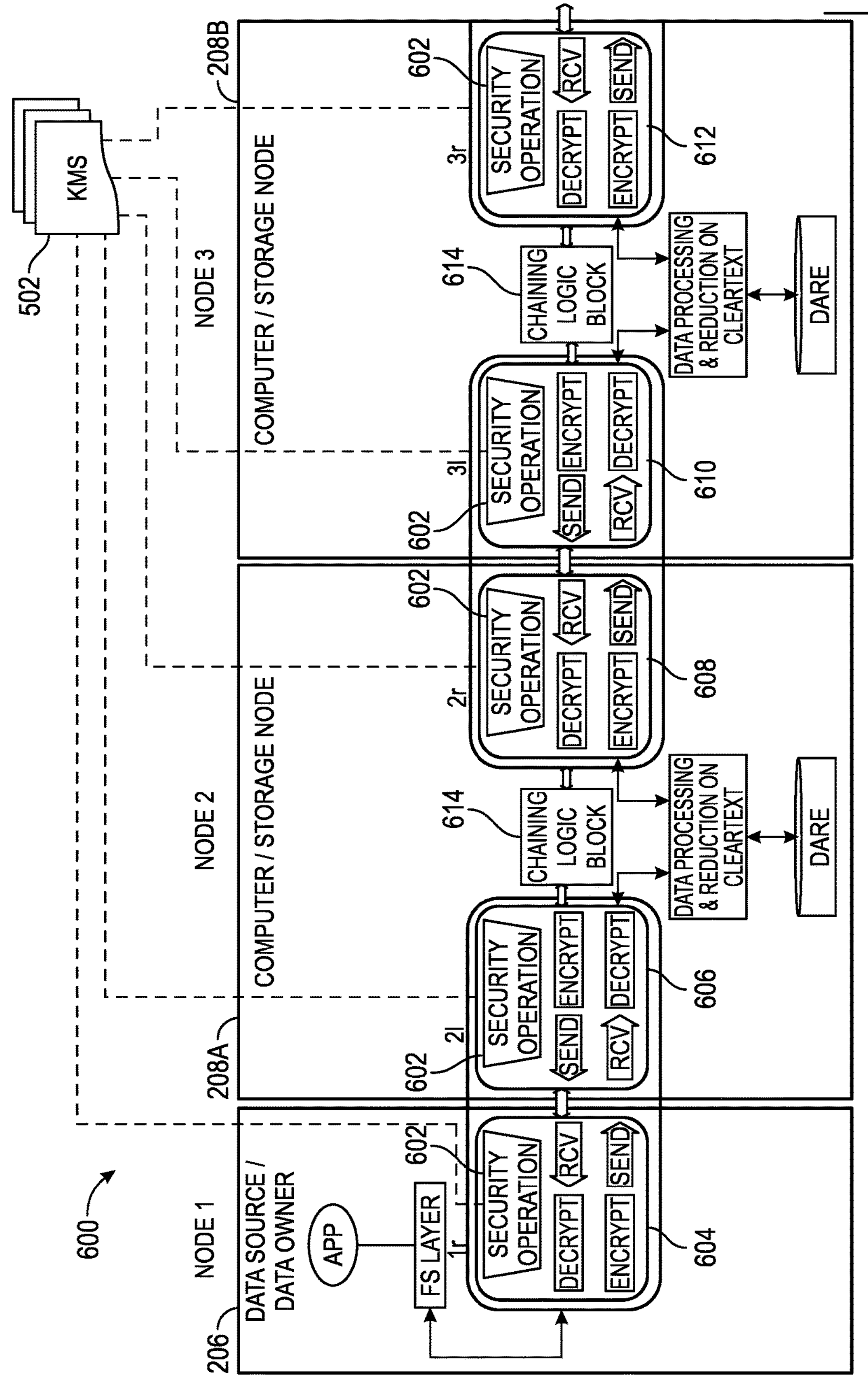

600
502
KMS
206
NODE 1
DATA SOURCE / DATA OWNER
APP
FS LAYER
1r
602
SECURITY OPERATION
DECRYPT
RCV
ENCRYPT
SEND
604
208A
NODE 2
COMPUTER / STORAGE NODE
2l
606
614
CHAINING LOGIC BLOCK
DATA PROCESSING & REDUCTION ON CLEARTEXT
DARE
2r
608
208B
NODE 3
3l
610
3r
612
202

FIG. 6

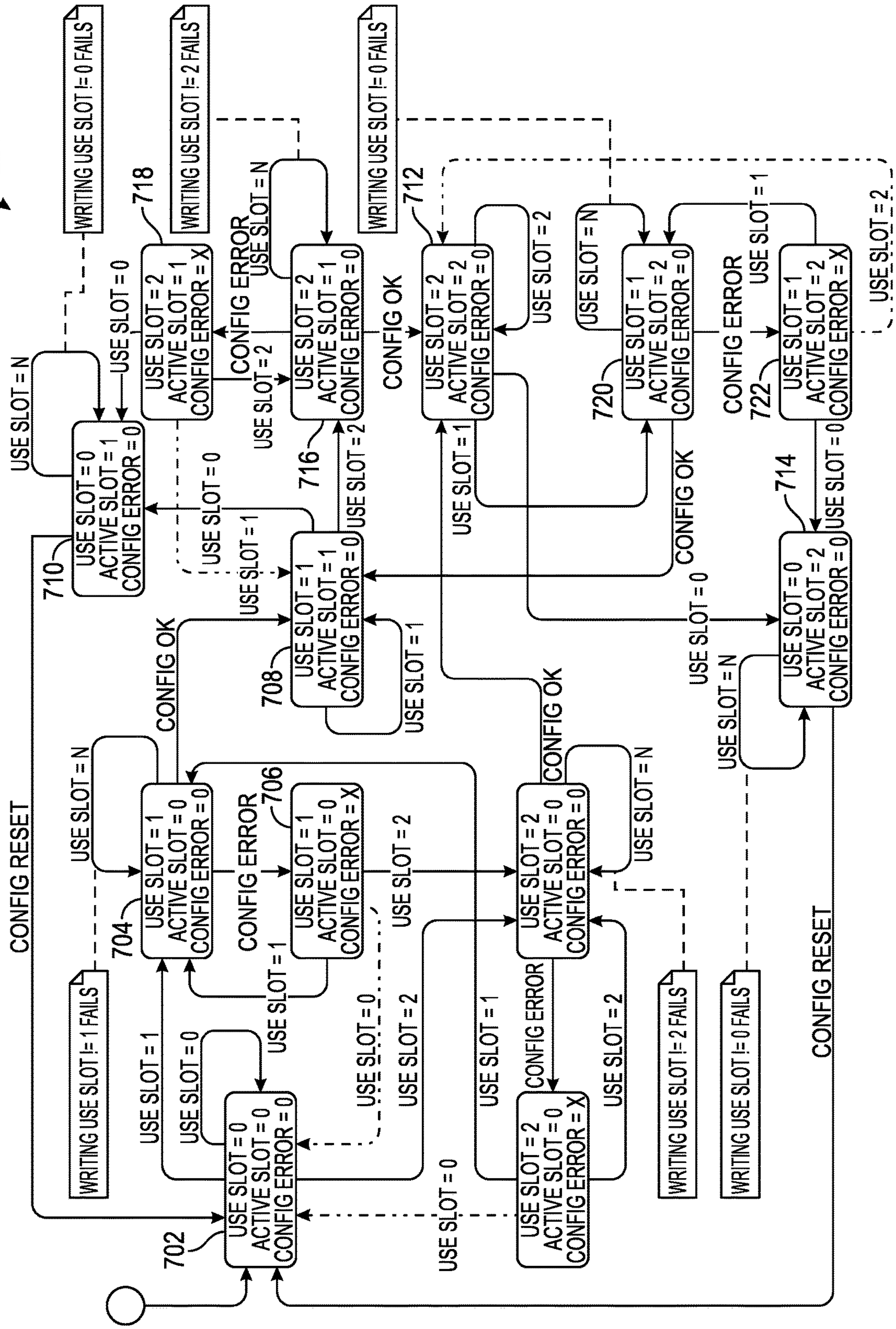

224
702
USE SLOT = 0
ACTIVE SLOT = 0
CONFIG ERROR = 0
704
USE SLOT = 1
ACTIVE SLOT = 0
CONFIG ERROR = 0
706
USE SLOT = 1
ACTIVE SLOT = 0
CONFIG ERROR = X
USE SLOT = 2
ACTIVE SLOT = 0
CONFIG ERROR = 0
USE SLOT = 2
ACTIVE SLOT = 0
CONFIG ERROR = X
708
USE SLOT = 1
ACTIVE SLOT = 1
CONFIG ERROR = 0
710
USE SLOT = 0
ACTIVE SLOT = 1
CONFIG ERROR = 0
712
USE SLOT = 2
ACTIVE SLOT = 2
CONFIG ERROR = 0
714
USE SLOT = 0
ACTIVE SLOT = 2
CONFIG ERROR = 0
716
USE SLOT = 2
ACTIVE SLOT = 1
CONFIG ERROR = 0
718
USE SLOT = 2
ACTIVE SLOT = 1
CONFIG ERROR = X
720
USE SLOT = 1
ACTIVE SLOT = 2
CONFIG ERROR = 0
722
USE SLOT = 1
ACTIVE SLOT = 2
CONFIG ERROR = X
WRITING USE SLOT != 0 FAILS
WRITING USE SLOT != 2 FAILS
WRITING USE SLOT != 0 FAILS
WRITING USE SLOT != 1 FAILS
WRITING USE SLOT != 2 FAILS
WRITING USE SLOT != 0 FAILS
CONFIG RESET
CONFIG RESET
CONFIG OK
CONFIG OK
CONFIG OK
CONFIG OK
CONFIG ERROR
CONFIG ERROR
CONFIG ERROR
CONFIG ERROR
USE SLOT = N
USE SLOT = 0
USE SLOT = 1
USE SLOT = 2

FIG. 7

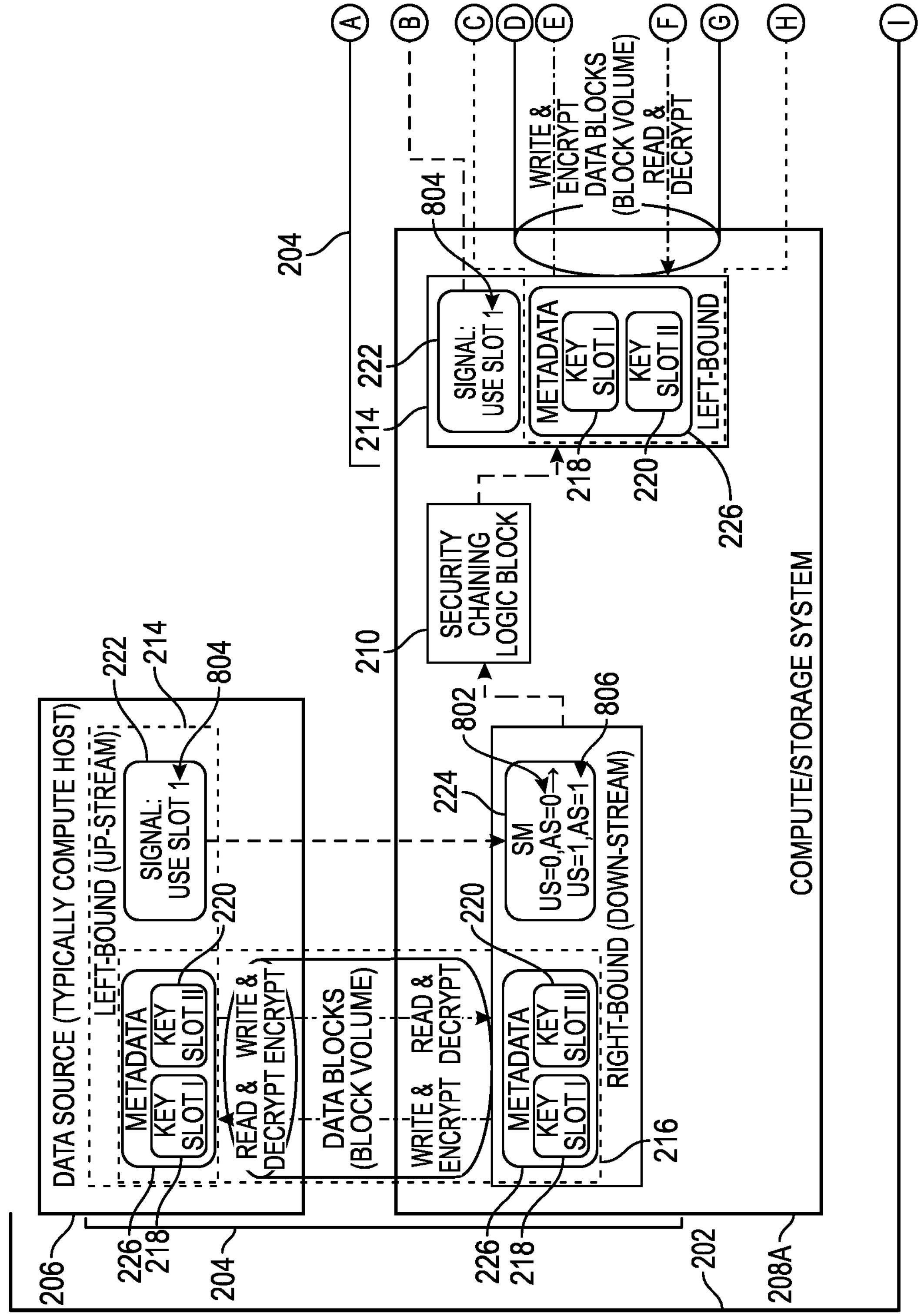

DATA SOURCE (TYPICALLY COMPUTE HOST)
LEFT-BOUND (UP-STREAM)
SIGNAL: USE SLOT 1
METADATA
KEY SLOT I
KEY SLOT II
READ & DECRYPT
WRITE & ENCRYPT
DATA BLOCKS (BLOCK VOLUME)
WRITE & ENCRYPT
READ & DECRYPT
RIGHT-BOUND (DOWN-STREAM)
SM
US=0,AS=0
US=1,AS=1
SECURITY CHAINING LOGIC BLOCK
SIGNAL: USE SLOT 1
LEFT-BOUND
WRITE & ENCRYPT
DATA BLOCKS (BLOCK VOLUME)
READ & DECRYPT
COMPUTE/STORAGE SYSTEM
A
B
C
D
E
F
G
H
I
202
204
206
208A
210
214
216
218
220
222
224
226
802
804
806

FIG. 8A

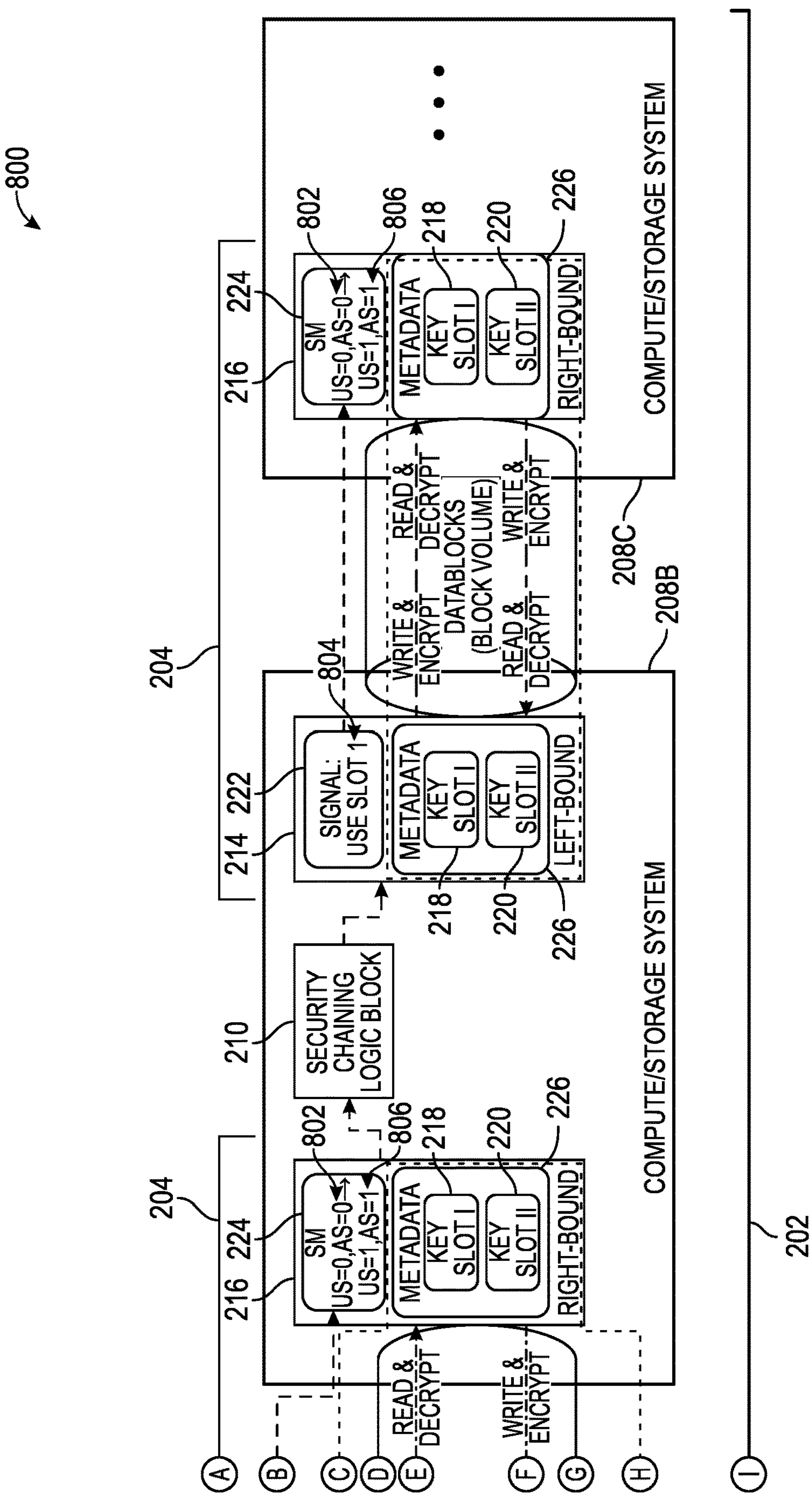

800
204
216
224
SM
US=0,AS=0
US=1,AS=1
802
806
METADATA
KEY SLOT I
KEY SLOT II
218
220
226
RIGHT-BOUND
210
SECURITY CHAINING LOGIC BLOCK
214
222
SIGNAL: USE SLOT 1
804
LEFT-BOUND
WRITE & ENCRYPT
READ & DECRYPT
DATABLOCKS (BLOCK VOLUME)
208B
208C
COMPUTE/STORAGE SYSTEM
202
A
B
C
D
E
F
G
H
I

FIG. 8B

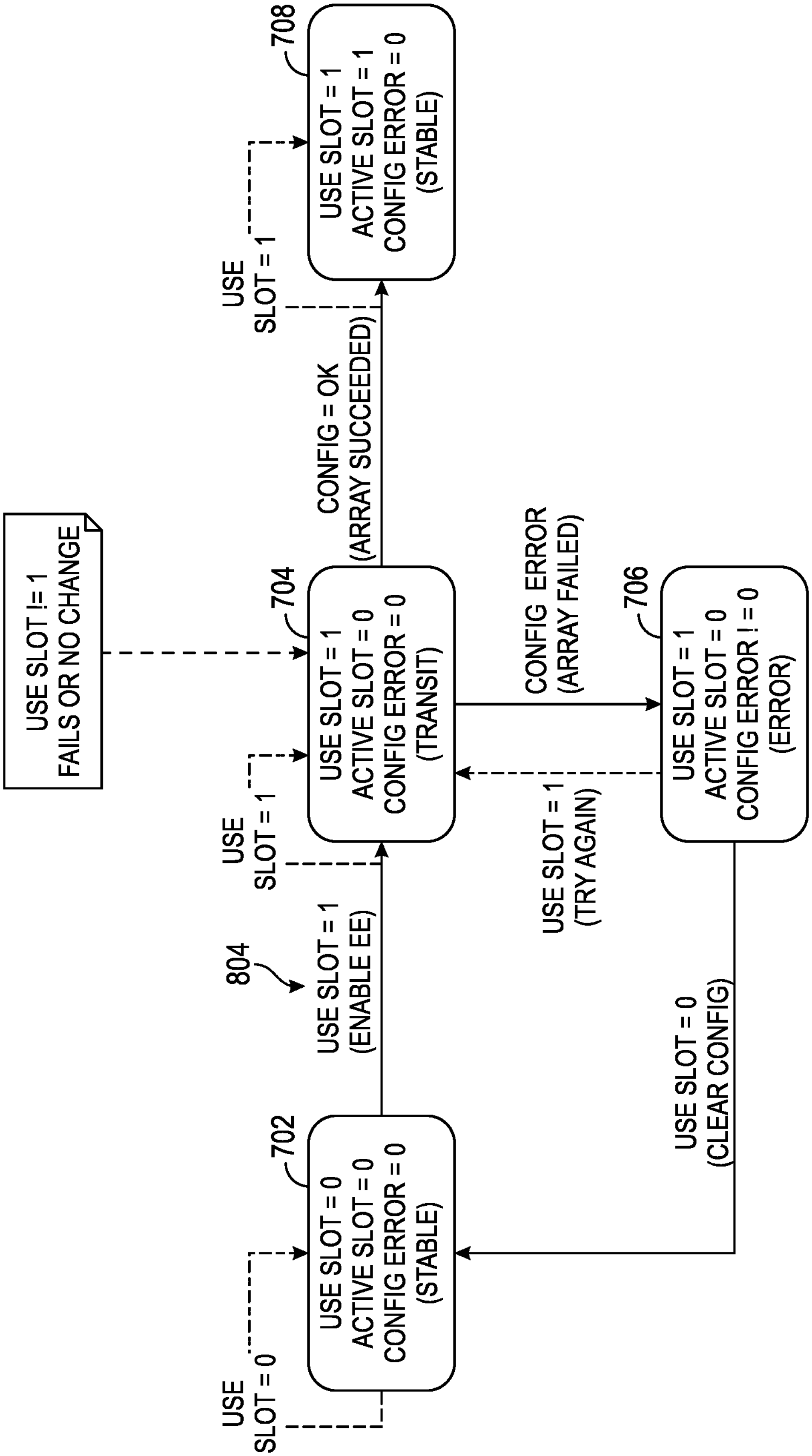

USE SLOT != 1
FAILS OR NO CHANGE
708
USE SLOT = 1
ACTIVE SLOT = 1
CONFIG ERROR = 0
(STABLE)
USE SLOT = 1
CONFIG = OK
(ARRAY SUCCEEDED)
704
USE SLOT = 1
ACTIVE SLOT = 0
CONFIG ERROR = 0
(TRANSIT)
USE SLOT = 1
CONFIG ERROR
(ARRAY FAILED)
706
USE SLOT = 1
ACTIVE SLOT = 0
CONFIG ERROR ! = 0
(ERROR)
USE SLOT = 1
(TRY AGAIN)
804
USE SLOT = 1
(ENABLE EE)
USE SLOT = 0
(CLEAR CONFIG)
702
USE SLOT = 0
ACTIVE SLOT = 0
CONFIG ERROR = 0
(STABLE)
USE SLOT = 0

FIG. 9

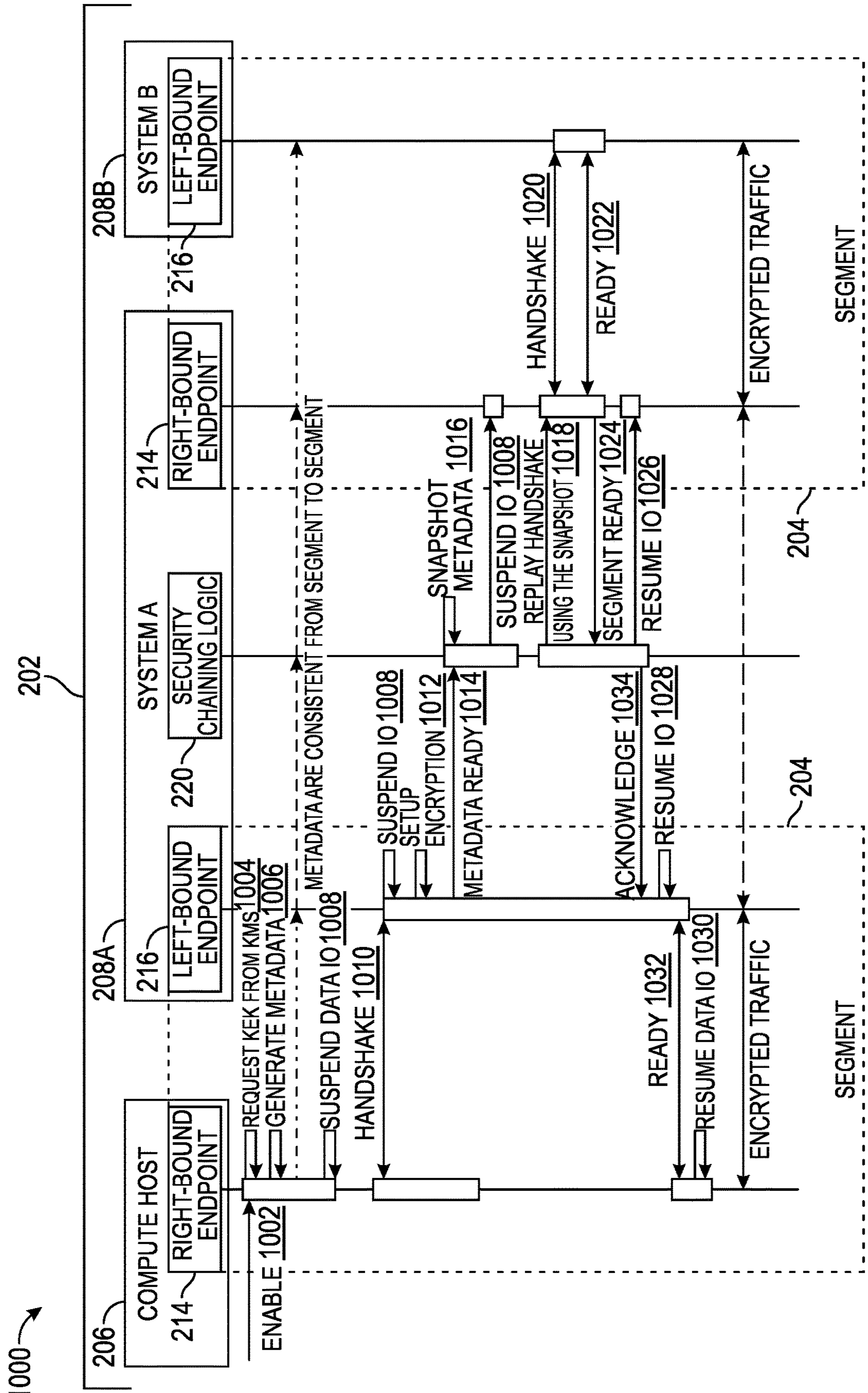
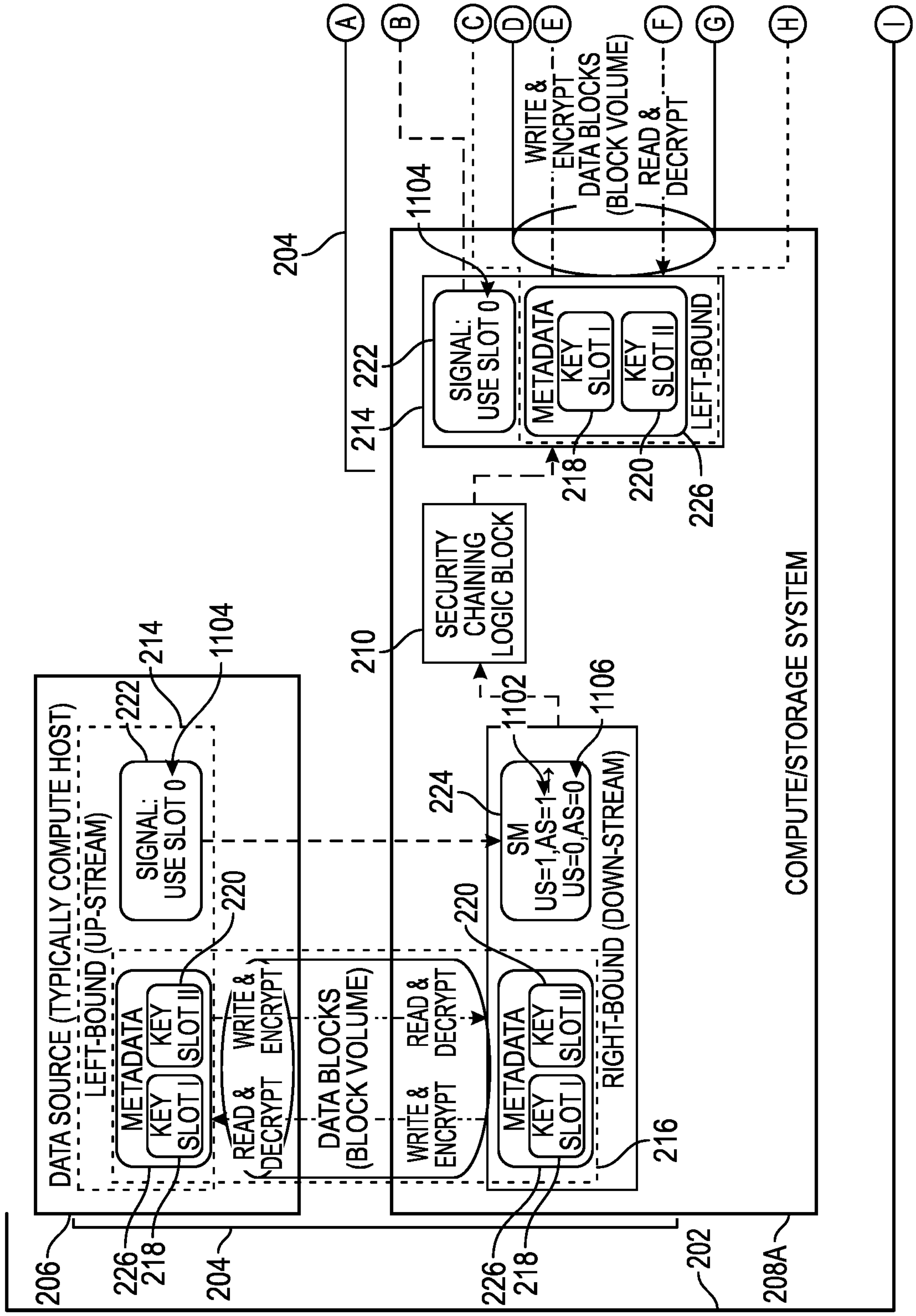

1000
202
206
COMPUTE HOST
214
RIGHT-BOUND ENDPOINT
208A
216
LEFT-BOUND ENDPOINT
SYSTEM A
220
SECURITY CHAINING LOGIC
214
RIGHT-BOUND ENDPOINT
208B
SYSTEM B
216
LEFT-BOUND ENDPOINT
ENABLE 1002
REQUEST KEK FROM KMS 1004
GENERATE METADATA 1006
SUSPEND DATA IO 1008
METADATA ARE CONSISTENT FROM SEGMENT TO SEGMENT
HANDSHAKE 1010
SUSPEND IO 1008
SETUP
ENCRYPTION 1012
METADATA READY 1014
SNAPSHOT METADATA 1016
SUSPEND IO 1008
REPLAY HANDSHAKE USING THE SNAPSHOT 1018
HANDSHAKE 1020
READY 1022
SEGMENT READY 1024
RESUME IO 1026
ACKNOWLEDGE 1034
RESUME IO 1028
READY 1032
RESUME DATA IO 1030
ENCRYPTED TRAFFIC
ENCRYPTED TRAFFIC
204
204
SEGMENT
SEGMENT

FIG. 10



A
B
C
D
E
F
G
H
I
204
1104
WRITE & ENCRYPT
DATA BLOCKS (BLOCK VOLUME)
READ & DECRYPT
214
222
SIGNAL: USE SLOT 0
METADATA
KEY SLOT I
KEY SLOT II
LEFT-BOUND
218
220
226
210
SECURITY CHAINING LOGIC BLOCK
1102
1106
224
SM
US=1,AS=1
US=0,AS=0
RIGHT-BOUND (DOWN-STREAM)
216
COMPUTE/STORAGE SYSTEM
DATA SOURCE (TYPICALLY COMPUTE HOST)
LEFT-BOUND (UP-STREAM)
206
202
208A

FIG. 11A

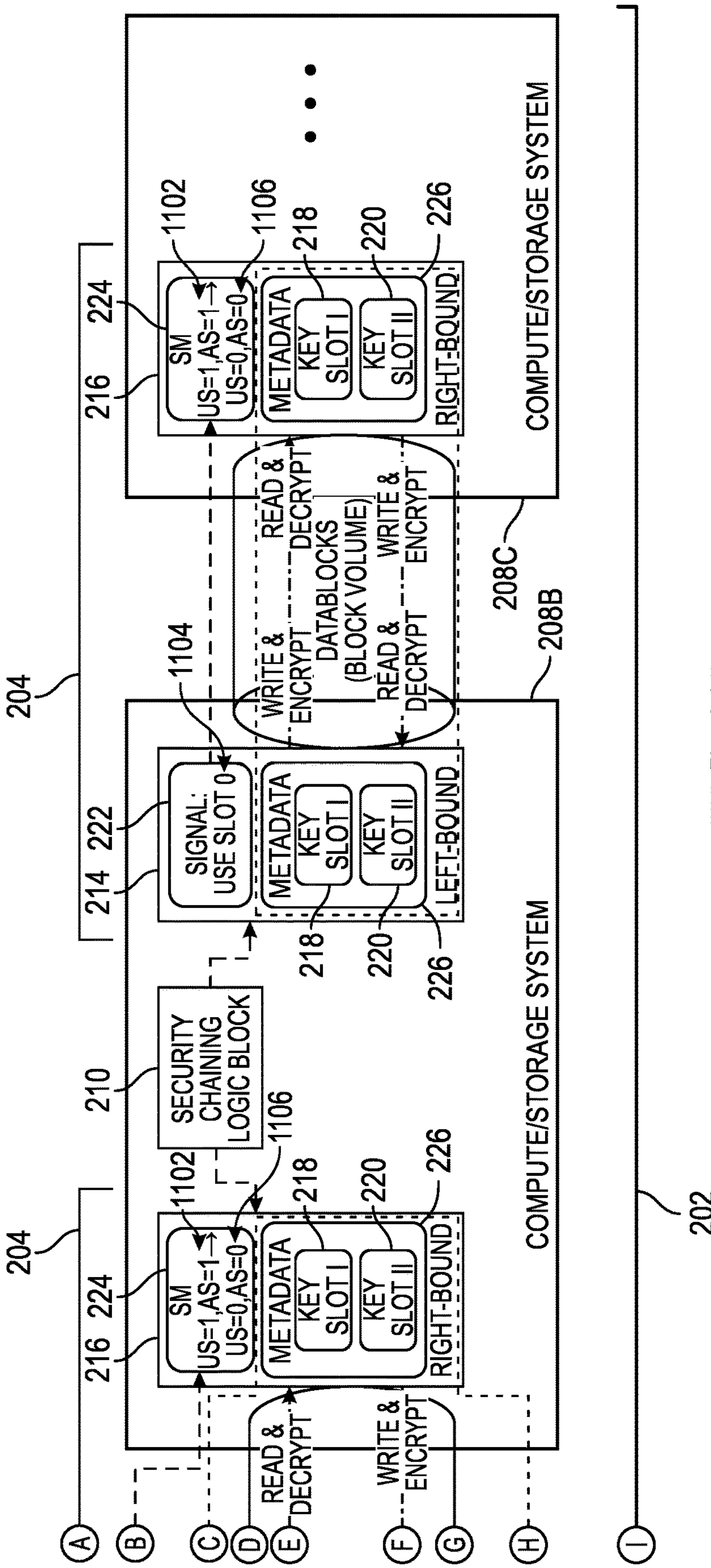

1100
204
216
224
SM
US=1,AS=1→
US=0,AS=0
1102
1106
METADATA
KEY SLOT I
KEY SLOT II
218
220
226
RIGHT-BOUND
COMPUTE/STORAGE SYSTEM
208C
208B
READ & DECRYPT
WRITE & ENCRYPT
DATABLOCKS (BLOCK VOLUME)
1104
214
222
SIGNAL: USE SLOT 0
LEFT-BOUND
210
SECURITY CHAINING LOGIC BLOCK
202
A
B
C
D
E
F
G
H
I

FIG. 11B

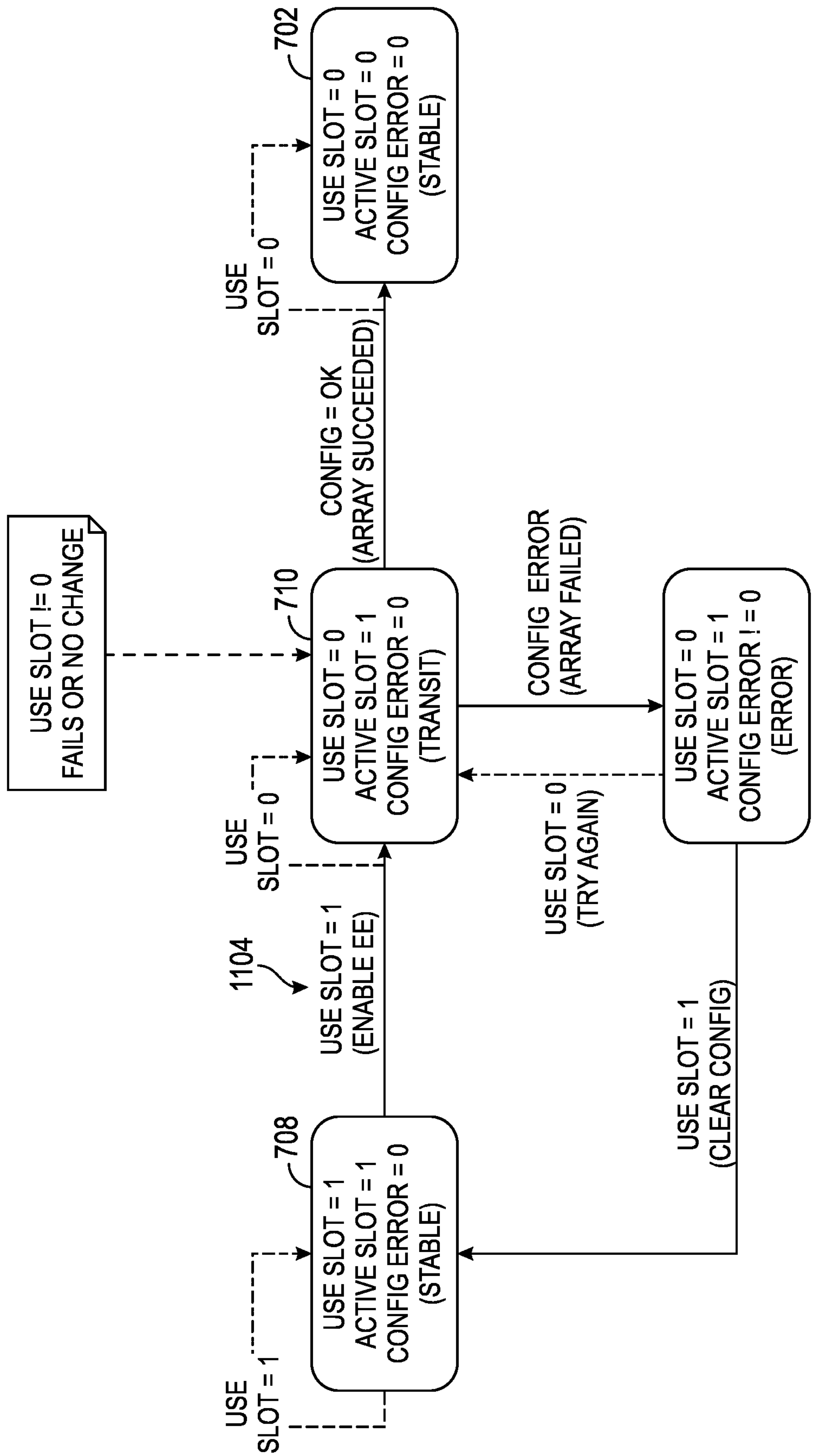

702
USE SLOT = 0
ACTIVE SLOT = 0
CONFIG ERROR = 0
(STABLE)
USE SLOT = 0
CONFIG = OK
(ARRAY SUCCEEDED)
USE SLOT != 0
FAILS OR NO CHANGE
710
USE SLOT = 0
ACTIVE SLOT = 1
CONFIG ERROR = 0
(TRANSIT)
USE SLOT = 0
CONFIG ERROR
(ARRAY FAILED)
USE SLOT = 0
ACTIVE SLOT = 1
CONFIG ERROR ! = 0
(ERROR)
USE SLOT = 0
(TRY AGAIN)
1104
USE SLOT = 1
(ENABLE EE)
USE SLOT = 1
(CLEAR CONFIG)
708
USE SLOT = 1
ACTIVE SLOT = 1
CONFIG ERROR = 0
(STABLE)
USE SLOT = 1

FIG. 12

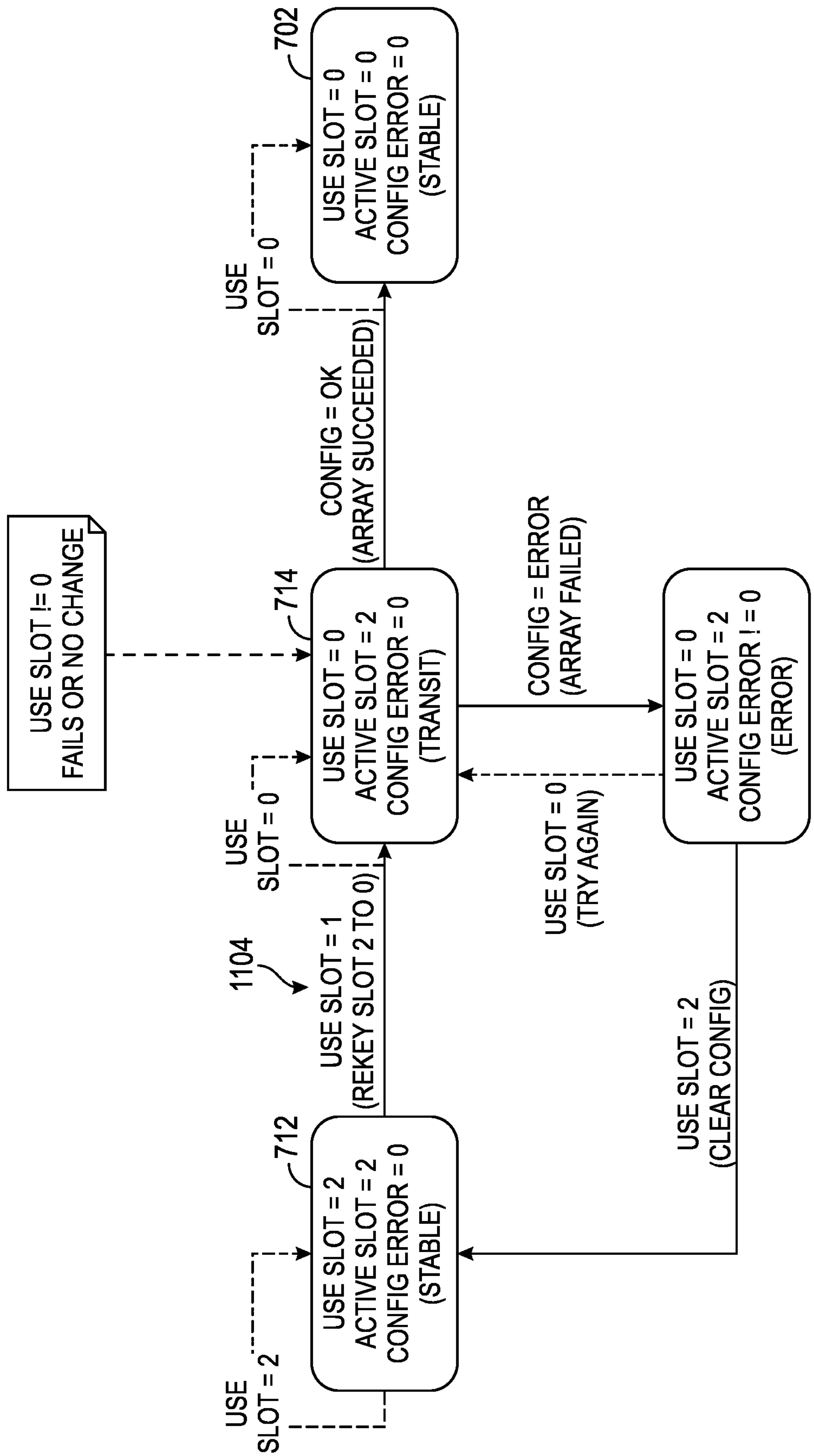

702
USE SLOT = 0
ACTIVE SLOT = 0
CONFIG ERROR = 0
(STABLE)
USE SLOT = 0
CONFIG = OK
(ARRAY SUCCEEDED)
USE SLOT != 0
FAILS OR NO CHANGE
714
USE SLOT = 0
ACTIVE SLOT = 2
CONFIG ERROR = 0
(TRANSIT)
USE SLOT = 0
CONFIG = ERROR
(ARRAY FAILED)
USE SLOT = 0
ACTIVE SLOT = 2
CONFIG ERROR ! = 0
(ERROR)
USE SLOT = 0
(TRY AGAIN)
1104
USE SLOT = 1
(REKEY SLOT 2 TO 0)
USE SLOT = 2
(CLEAR CONFIG)
712
USE SLOT = 2
ACTIVE SLOT = 2
CONFIG ERROR = 0
(STABLE)
USE SLOT = 2

FIG. 13

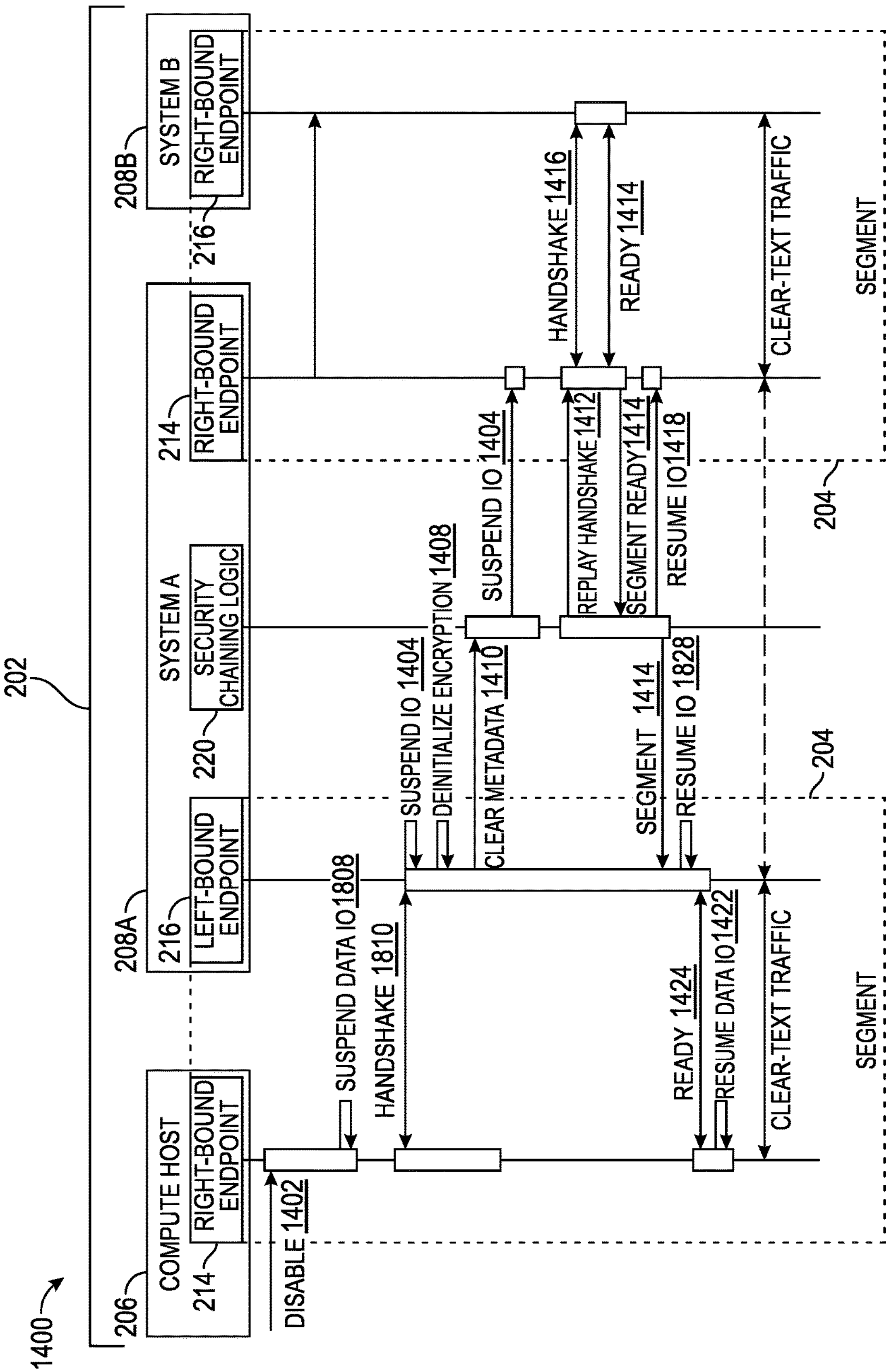
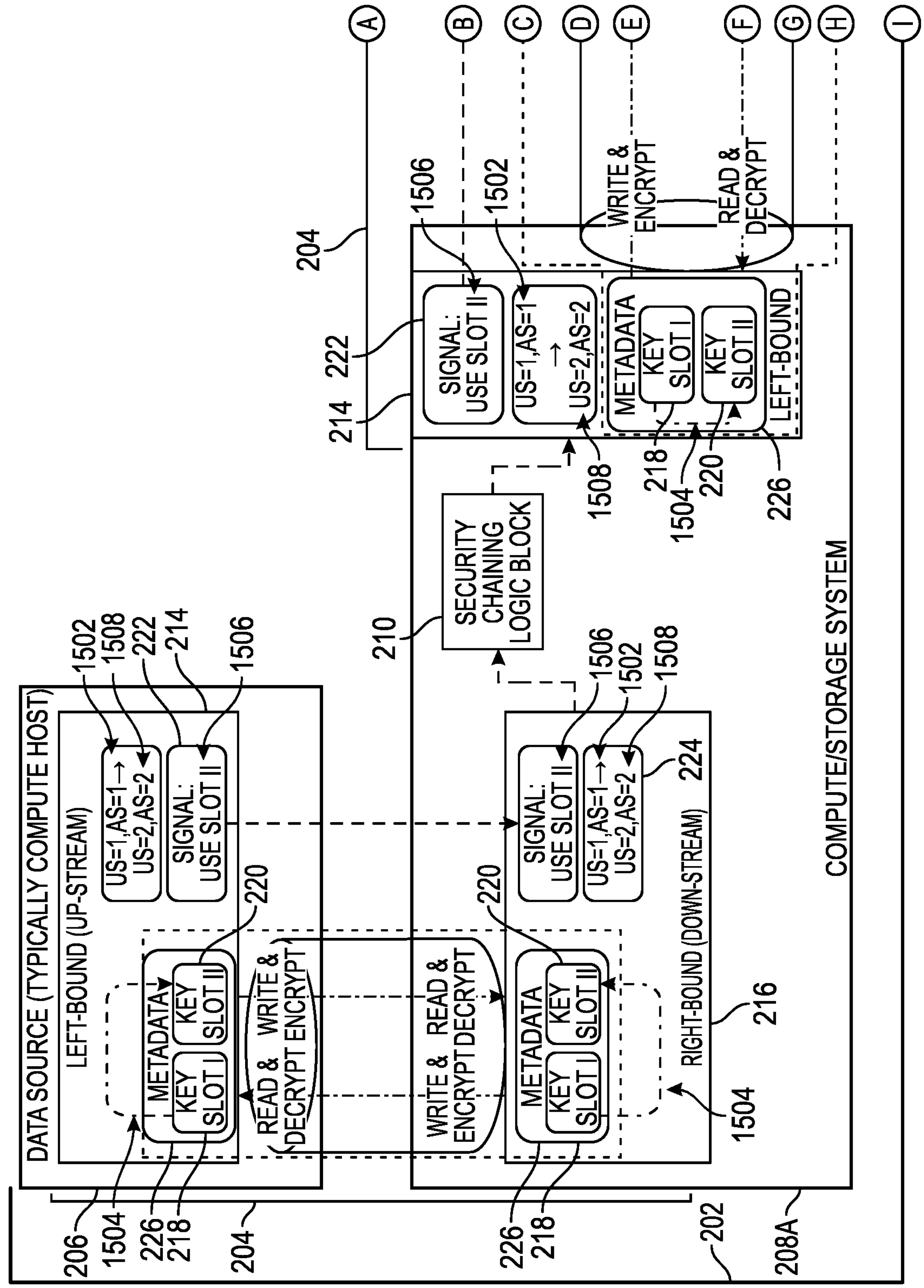

1400
202
206
COMPUTE HOST
214
RIGHT-BOUND ENDPOINT
208A
216
LEFT-BOUND ENDPOINT
SYSTEM A
220
SECURITY CHAINING LOGIC
214
RIGHT-BOUND ENDPOINT
208B
SYSTEM B
216
RIGHT-BOUND ENDPOINT
DISABLE 1402
SUSPEND DATA IO 1808
HANDSHAKE 1810
SUSPEND IO 1404
DEINITIALIZE ENCRYPTION 1408
CLEAR METADATA 1410
SUSPEND IO 1404
REPLAY HANDSHAKE 1412
SEGMENT READY 1414
RESUME IO 1418
SEGMENT 1414
RESUME IO 1828
READY 1424
RESUME DATA IO 1422
HANDSHAKE 1416
READY 1414
CLEAR-TEXT TRAFFIC
CLEAR-TEXT TRAFFIC
SEGMENT
SEGMENT
204
204

FIG. 14



DATA SOURCE (TYPICALLY COMPUTE HOST)
LEFT-BOUND (UP-STREAM)
US=1,AS=1→
US=2,AS=2
SIGNAL:
USE SLOT II
METADATA
KEY
SLOT I
KEY
SLOT II
READ &
DECRYPT
WRITE &
ENCRYPT
WRITE &
ENCRYPT
READ &
DECRYPT
METADATA
KEY
SLOT I
KEY
SLOT II
RIGHT-BOUND (DOWN-STREAM)
SIGNAL:
USE SLOT II
US=1,AS=1→
US=2,AS=2
SECURITY
CHAINING
LOGIC BLOCK
SIGNAL:
USE SLOT II
US=1,AS=1
→
US=2,AS=2
METADATA
KEY
SLOT I
KEY
SLOT II
LEFT-BOUND
WRITE &
ENCRYPT
READ &
DECRYPT
COMPUTE/STORAGE SYSTEM
A
B
C
D
E
F
G
H
I
202
204
206
208A
210
214
216
218
220
222
224
226
1502
1504
1506
1508

FIG. 15A

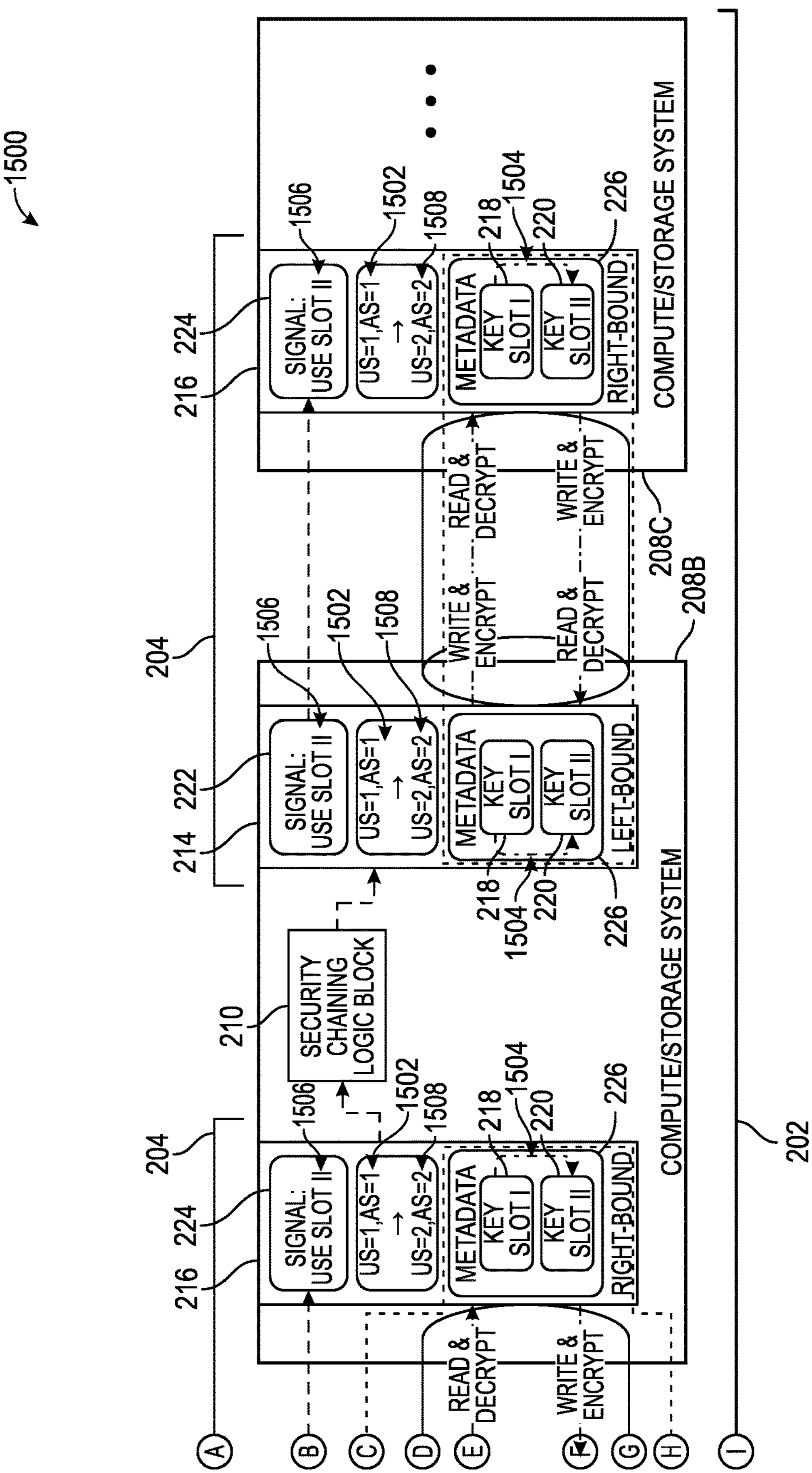

1500
SIGNAL: USE SLOT II
US=1,AS=1
→
US=2,AS=2
METADATA
KEY SLOT I
KEY SLOT II
RIGHT-BOUND
LEFT-BOUND
SECURITY CHAINING LOGIC BLOCK
COMPUTE/STORAGE SYSTEM
READ & DECRYPT
WRITE & ENCRYPT
202
204
208B
208C
210
214
216
218
220
222
224
226
1502
1504
1506
1508
A
B
C
D
E
F
G
H
I

FIG. 15B

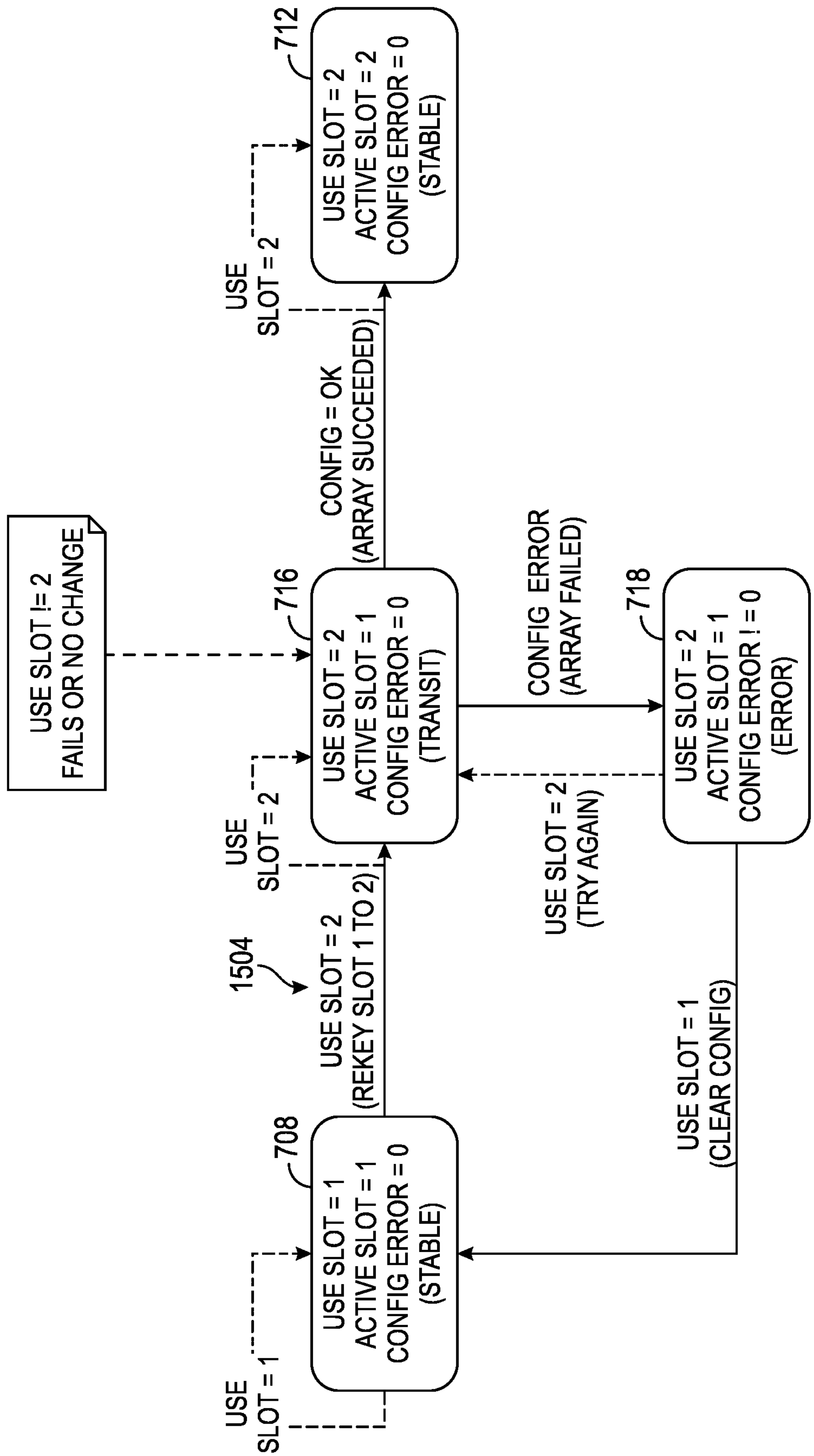
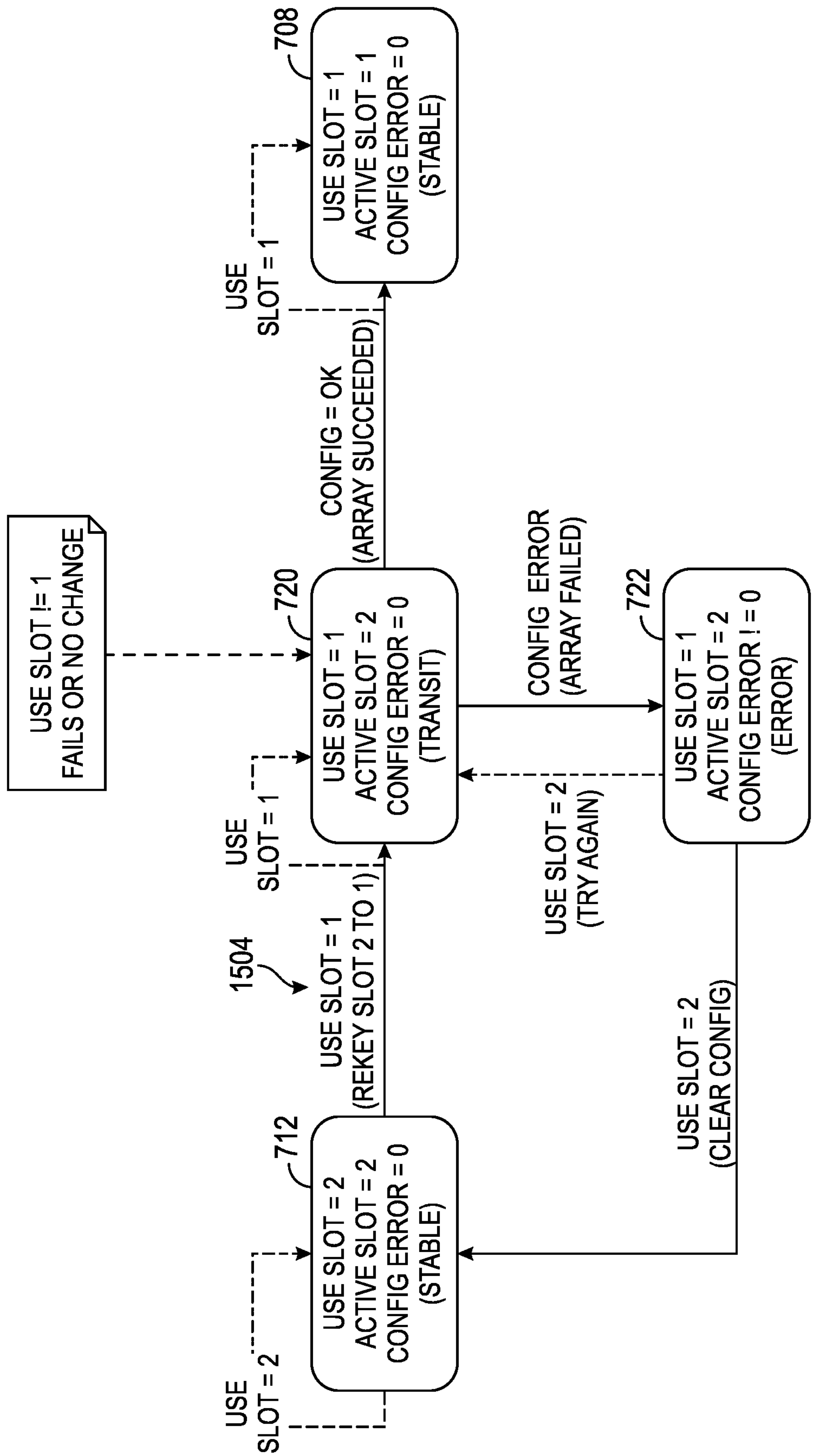

USE SLOT != 2
FAILS OR NO CHANGE
708
USE SLOT = 1
ACTIVE SLOT = 1
CONFIG ERROR = 0
(STABLE)
USE
SLOT = 1
1504
USE SLOT = 2
(REKEY SLOT 1 TO 2)
716
USE SLOT = 2
ACTIVE SLOT = 1
CONFIG ERROR = 0
(TRANSIT)
USE
SLOT = 2
CONFIG = OK
(ARRAY SUCCEEDED)
712
USE SLOT = 2
ACTIVE SLOT = 2
CONFIG ERROR = 0
(STABLE)
USE
SLOT = 2
CONFIG ERROR
(ARRAY FAILED)
718
USE SLOT = 2
ACTIVE SLOT = 1
CONFIG ERROR ! = 0
(ERROR)
USE SLOT = 2
(TRY AGAIN)
USE SLOT = 1
(CLEAR CONFIG)

FIG. 16



708
USE SLOT = 1
ACTIVE SLOT = 1
CONFIG ERROR = 0
(STABLE)
USE SLOT = 1
CONFIG = OK
(ARRAY SUCCEEDED)
USE SLOT != 1
FAILS OR NO CHANGE
720
USE SLOT = 1
ACTIVE SLOT = 2
CONFIG ERROR = 0
(TRANSIT)
USE SLOT = 1
CONFIG ERROR
(ARRAY FAILED)
722
USE SLOT = 1
ACTIVE SLOT = 2
CONFIG ERROR ! = 0
(ERROR)
USE SLOT = 2
(TRY AGAIN)
1504
USE SLOT = 1
(REKEY SLOT 2 TO 1)
USE SLOT = 2
(CLEAR CONFIG)
712
USE SLOT = 2
ACTIVE SLOT = 2
CONFIG ERROR = 0
(STABLE)
USE SLOT = 2

FIG. 17

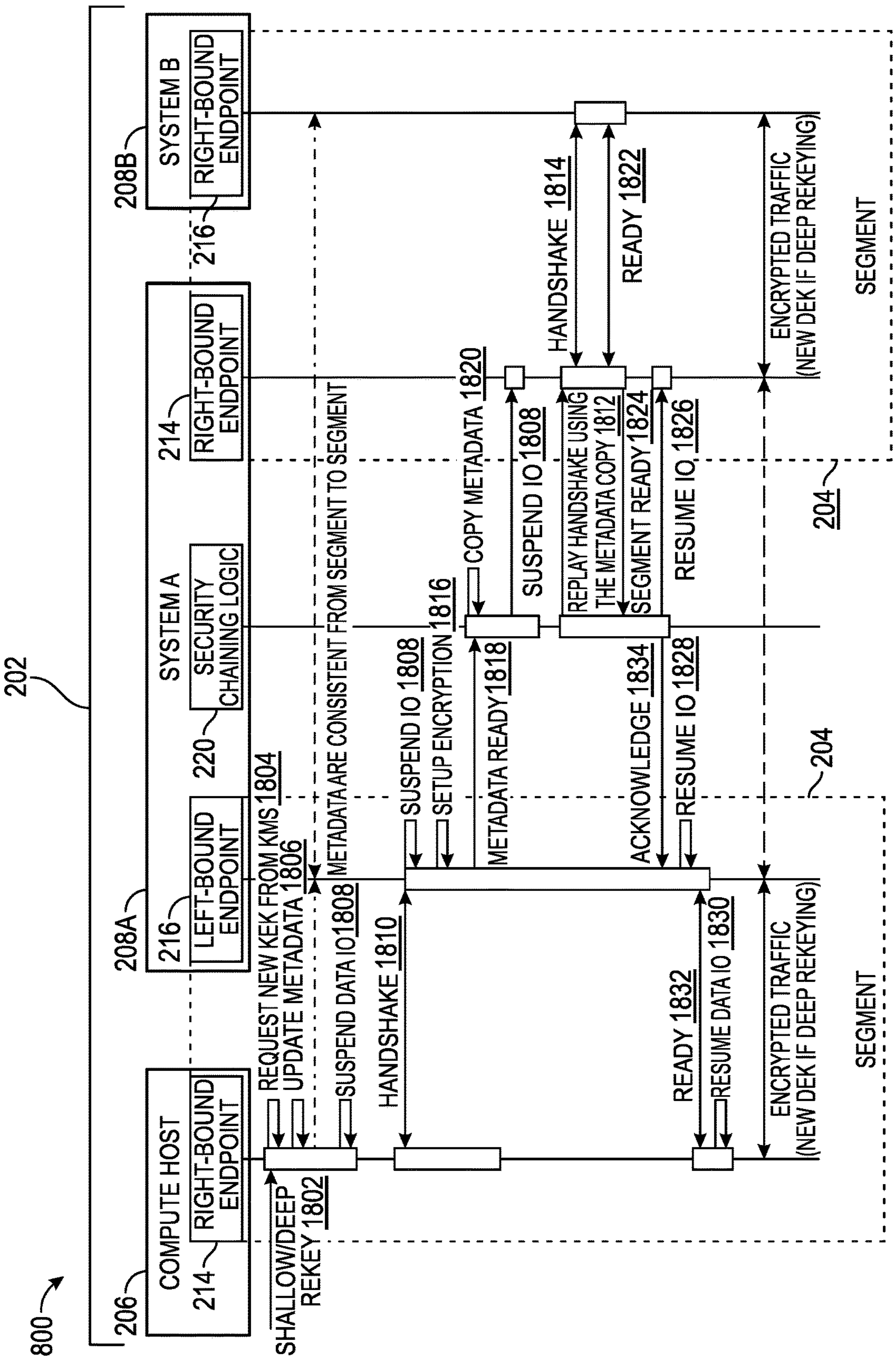

1800
202
206
COMPUTE HOST
214
RIGHT-BOUND ENDPOINT
208A
216
LEFT-BOUND ENDPOINT
220
SECURITY CHAINING LOGIC
SYSTEM A
214
RIGHT-BOUND ENDPOINT
208B
SYSTEM B
216
RIGHT-BOUND ENDPOINT
SHALLOW/DEEP REKEY 1802
REQUEST NEW KEK FROM KMS 1804
UPDATE METADATA 1806
SUSPEND DATA IO 1808
METADATA ARE CONSISTENT FROM SEGMENT TO SEGMENT
HANDSHAKE 1810
SUSPEND IO 1808
SETUP ENCRYPTION 1816
METADATA READY 1818
COPY METADATA 1820
SUSPEND IO 1808
REPLAY HANDSHAKE USING THE METADATA COPY 1812
SEGMENT READY 1824
ACKNOWLEDGE 1834
RESUME IO 1828
RESUME IO 1826
HANDSHAKE 1814
READY 1822
READY 1832
RESUME DATA IO 1830
ENCRYPTED TRAFFIC (NEW DEK IF DEEP REKEYING)
ENCRYPTED TRAFFIC (NEW DEK IF DEEP REKEYING)
SEGMENT
SEGMENT
204
204

FIG. 18

REKEYING END-TO-END EFFICIENT ENCRYPTION WITH SECURITY CHAINING

CROSS-REFERENCE TO RELATED APPLICATIONS

This disclosure is related to U.S. patent application Ser. No., entitled End-to-End Efficient Encryption with Security Chaining, filed concurrent herewith, on Jul. 29, 2022, and to U.S. patent application Ser. No., entitled Enablement and Disablement of End-to-End Efficient Encryption with Security Chaining, also filed concurrent herewith, on Jul. 29, 2022, both of which are incorporated herein by reference.

FIELD

This disclosure relates generally to IHSs (IHSs), and, more specifically, to systems and methods of for rekeying end-to-end efficient encryption with security chaining.

BACKGROUND

As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is Information Handling Systems (IHSs). An IHS generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, IHSs may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in IHSs allow for IHSs to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, IHSs may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

Data, from source to destination, have many segments in a network path. Typical end-to-end encryption deals with one segment of the network path and are lacking in security coordination of multiple segments of the complete network path. Typical end-to-end encryption also lacks operational capability for online and/or instant rekey (e.g., refreshing the encryption key of an existing encrypted system, such as peer-to-peer encrypted communication, or data on a storage medium that is encrypted) and existing data sets encryption and other end-to-end security control flexibilities.

SUMMARY

Embodiments of end-to-end efficient encryption with security chaining are described. In an illustrative, non-limiting example rekeying an Information Handling System (IHS) network End-to-End Efficient Encryption (E2EEE) with security chaining includes locking an encrypted data volume, preventing reading of, and writing to, the encrypted data volume by applications. A data source IHS may request a new key from a key management system and write new metadata in a trailer of the encrypted data block using a different key slot than a currently used and active metadata key slot, wherein the different key slot is updated with the with the new key. A security chaining logic component in each network IHS, may maintain the new metadata across all data connection segment interfaces of the network IHSs.

The data source IHS sends an out-of-data signal to change the use key slot from the currently used key slot to the different key slot. An IHS security chaining logic component in each network IHS may regenerate this out-of-data signal to change the use key slot from the currently used key slot to the different key slot to each next IHS E2EEE data connection segment interface.

In response to the out-of-data signal to change the use key slot from the currently used key slot to the different key slot, the encryption configuration state machine of each IHS E2EEE data connection segment interface changes the state machine state from the currently used and active metadata key slot to use the different metadata key slot and make the different metadata key slot the active key slot. This may include transiting, in the encryption configuration state machine of each data connection segment interface, in response to the out-of-data signal to change the use key slot from the currently used key slot to the different key slot, a state with the currently used and active metadata key slot to use the different metadata key slot, but maintain the currently active metadata key slot, prior to making the different metadata key slot the active key slot, maintaining encryption and decryption prior to unlocking the encrypted data volume occurring on all data in the network using the currently active key slot.

In various implementations, the security chaining logic component in each network IHS may enable changing the encryption configuration state machine of each data connection segment interface to a state to use the different metadata key slot and making the different metadata key slot the active key slot, backward from a last data connection segment interface to a first data connection segment interface, such that the last data connection segment interface to the first data connection segment interface progressively switch to the different key slot.

The data source IHS may then unlock the encrypted data volume, enabling writing and/or reading user data by the data source IHS and encryption and decryption in all IHS E2EEE data connection segment interfaces. However, prior to unlocking the encrypted data volume, a state machine of each IHS E2EEE data connection segment interface may be maintained in a state with the currently used and active metadata key slot, maintaining encryption and decryption prior to unlocking the encrypted data volume occurring on all data in the network using the currently used key slot. Also, in various implementations, in response to a first data connection segment interface receiving a metadata update, the security chaining logic component, may interrupt and roll back an out-of-data signal to change the use key slot from the currently used key slot to the different key slot to a second data connection segment interface, and wait for a ready state in the metadata on the second data connection segment interface to unlock the encrypted data volume.

In various implementations, the security chaining logic component in each network IHS may enable encryption and decryption in all IHS E2EEE data connection segment interfaces forward from a first data connection segment interface to a last data connection segment interface, such that encryption and decryption in all IHS E2EEE data connection segment interfaces use the different key slot.

This rekeying may be shallow, rekeying a key encryption key and any intermediate key, or this rekeying may be deep, rekeying a key encryption key, any intermediate key or keys and a data encryption key.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention(s) is/are illustrated by way of example and is/are not limited by the accompanying figures, in which like references indicate similar elements. Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale.

FIG. **1** is a block diagram illustrating components of an example of an Information Handling System (IHS), according to some embodiments.

FIGS. **2**A and **2**B are partial views intended to form one complete view of a block diagram of an IHS network environment showing end-to-end efficient encryption with security chaining according to some embodiments.

FIGS. **3**A and **3**B are partial views intended to form one complete view of a block diagram of an IHS network environment showing multiple parallel end-to-end efficient encryption with security chaining of multiple segment chains, according to some embodiments.

FIG. **4** is a block diagram of an example end-to-end efficient encryption segment, according to some embodiments.

FIG. **5** is a block diagram of the example end-to-end efficient encryption segment of FIG. **4**, showing communications within the segment, according to some embodiments.

FIG. **6** is a block workflow diagram of end-to-end efficient encryption security chaining, according to some embodiments.

FIG. **7** is a diagram of a (complete) compute host and/or storage IHS state machine for end-to-end efficient encryption with security chaining, according to some embodiments.

FIGS. **8**A and **8**B are partial views intended to form one complete view of a block diagram of an IHS network environment showing enablement of end-to-end efficient encryption with security chaining.

FIG. **9** is a state machine diagram for enablement of end-to-end efficient encryption with security chaining, according to some embodiments.

FIG. **10** is a swim lane workflow diagram showing enablement of end-to-end efficient encryption with security chaining, according to some embodiments.

FIGS. **11**A and **11**B are partial views intended to form one complete view of a block diagram of an IHS network environment showing disablement of end-to-end efficient encryption with security chaining, according to some embodiments

FIG. **12** is a state machine diagram for disablement of end-to-end efficient encryption with security chaining, according to some embodiments.

FIG. **13** is another state machine diagram for disablement of end-to-end efficient encryption with security chaining, according to some embodiments.

FIG. **14** is a swim lane workflow diagram showing disablement of end-to-end efficient encryption with security chaining, according to some embodiments.

FIGS. **15**A and **15**B are partial views intended to form one complete view of a block diagram of an IHS network environment showing rekeying in end-to-end efficient encryption with security chaining, according to some embodiments.

FIG. **16** is a state machine diagram for rekeying in end-to-end efficient encryption with security chaining, according to some embodiments.

FIG. **17** is another state machine diagram for rekeying in end-to-end efficient encryption with security chaining, according to some embodiments.

FIG. **18** is a swim lane workflow diagram showing rekeying in end-to-end efficient encryption with security chaining, according to some embodiments.

DETAILED DESCRIPTION

For purposes of this disclosure, an Information Handling System (IHS) may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an IHS may, in accordance with embodiments of the present systems and methods, be a server (e.g., compute sled, storage sled, blade server, rack server, etc.), a network storage device, a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), or any other suitable device and may vary in size, shape, performance, functionality, and price. The IHS may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the IHS may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The IHS may also include one or more buses operable to transmit communications between the various hardware components.

A more detailed example of an IHS is described with respect to FIG. **1**, in that FIG. **1** is a block diagram illustrating components of example IHS **100**, according to some embodiments. IHS **100** may utilize one or more processors **105**. In some embodiments, processors **105** may include a main processor and a co-processor, each of which may include a plurality of processing cores that, in certain scenarios, may each be used to run an instance of a server process. In certain embodiments, one or all of processor(s) **105** may be graphics processing units (GPUs) in scenarios where IHS **100** has been configured to support functions such as multimedia services and graphics applications.

As illustrated, processor(s) **105** includes an integrated memory controller **110** that may be implemented directly within the circuitry of the processor **105**, or the memory controller **110** may be a separate integrated circuit that is located on the same die as the processor **105**. The memory controller **110** may be configured to manage the transfer of data to and from the system memory **115** of the IHS via a high-speed memory interface **120**. The system memory **115** is coupled to processor(s) **105** via a memory bus **120** that provides the processor(s) **105** with high-speed memory used in the execution of computer program instructions by the processor(s) **105**. Accordingly, system memory **115** may include memory components, such as static RAM (SRAM), dynamic RAM (DRAM), NAND Flash memory, suitable for supporting high-speed memory operations by the processor(s) **105**. In certain embodiments, system memory **115** may combine both persistent, non-volatile memory and volatile memory.

In certain embodiments, the system memory **115** may be comprised of multiple removable memory modules. The system memory **115** of the illustrated embodiment includes removable memory modules **115***a-n*. Each of the removable memory modules **115***a-n* may correspond to a printed circuit board memory socket that receives a removable memory module **115***a-n*, such as a Dual In-line Memory Module (DIMM), that can be coupled to the socket and then decoupled from the socket as needed, such as to upgrade memory capabilities or to replace faulty memory modules. Other embodiments of IHS memory **115** may be configured with memory socket interfaces that correspond to different types of removable memory module form factors, such as a Dual In-line Package (DIP) memory, a Single In-line Pin Package (SIPP) memory, a Single In-line Memory Module (SIMM), and/or a Ball Grid Array (BGA) memory, or soldered-in memory modules affixed to an IHS motherboard, etc.

IHS **100** may utilize chipset **125** that may be implemented by integrated circuits that are coupled to processor(s) **105**. In this embodiment, processor(s) **105** is depicted as a component of chipset **125**. In other embodiments, all of chipset **125**, or portions of chipset **125** may be implemented directly within the integrated circuitry of processor(s) **105**. The chipset may provide the processor(s) **105** with access to a variety of resources accessible via one or more buses **130**. Various embodiments may utilize any number of buses to provide the illustrated pathways served by bus **130**. In certain embodiments, bus **130** may include a PCIe switch fabric that is accessed via a PCIe root complex.

As illustrated, IHS **100** includes BMC **135** to provide capabilities for remote monitoring and management of various aspects of IHS **100**. In support of these operations, BMC **135** may utilize both in-band, sideband and/or out-of-band communications with certain managed components of IHS **100**, such as, for example, processor(s) **105**, system memory **115**, chipset **125**, network controller **140**, storage device(s) **145**, etc. BMC **135** may be installed on the motherboard of IHS **100** or may be coupled to IHS **100** via an expansion slot provided by the motherboard. As a non-limiting example of a BMC, the integrated Dell Remote Access Controller (iDRAC) from Dell® is embedded within Dell PowerEdge™ servers and provides functionality that helps information technology (IT) administrators deploy, update, monitor, and maintain servers remotely. BMC **135** may include non-volatile memory having program instructions stored thereon that are usable by CPU(s) **105** to enable remote management of IHS **100**. For example, BMC **135** may enable a user to discover, configure, and manage BMC **135**, setup configuration options, resolve and administer hardware or software problems, etc. Additionally, or alternatively, BMC **135** may include one or more firmware volumes, each volume having one or more firmware files used by the BIOS' firmware interface to initialize and test components of IHS **100**.

IHS **100** may also include the one or more I/O ports **150**, such as USB ports, PCIe ports, TPM (Trusted Platform Module) connection ports, HDMI ports, audio ports, docking ports, network ports, Fibre Channel ports and other storage device ports. Such I/O ports **150** may be externally accessible or may be internal ports that are accessed by opening the enclosure of the IHS **100**. Through couplings made to these I/O ports **150**, users may couple the IHS **100** directly to other IHSs, storage resources, external networks and a vast variety of peripheral components.

As illustrated, IHS **100** may include one or more FPGA (Field-Programmable Gate Array) cards **155**. Each of the FPGA card **155** supported by IHS **100** may include various processing and memory resources, in addition to an FPGA logic unit that may include circuits that can be reconfigured after deployment of IHS **100** through programming functions supported by the FPGA card **155**. Through such reprogramming of such logic units, each individual FGPA card **155** may be optimized to perform specific processing tasks, such as specific signal processing, security, data mining, and artificial intelligence functions, and/or to support specific hardware coupled to IHS **100**. In some embodiments, a single FPGA card **155** may include multiple FPGA logic units, each of which may be separately programmed to implement different computing operations, such as in computing different operations that are being offloaded from processor **105**.

IHS **100** may include one or more storage controllers **160** that may be utilized to access storage devices **145***a-n* that are accessible via the chassis in which IHS **100** is installed. Storage controller **160** may provide support for RAID (Redundant Array of Independent Disks) configurations of logical and physical storage devices **145***a-n*. In some embodiments, storage controller **160** may be an HBA (Host Bus Adapter) that provides more limited capabilities in accessing physical storage devices **145***a-n*. In some embodiments, storage devices **145***a-n* may be replaceable, hot-swappable storage devices that are installed within bays provided by the chassis in which IHS **100** is installed. In embodiments where storage devices **145***a-n* are hot-swappable devices that are received by bays of chassis, the storage devices **145***a-n* may be coupled to IHS **100** via couplings between the bays of the chassis and a midplane of IHS **100**. In some embodiments, storage devices **145***a-n* may also be accessed by other IHSs that are also installed within the same chassis as IHS **100**. Storage devices **145***a-n* may include SAS (Serial Attached SCSI) magnetic disk drives, SATA (Serial Advanced Technology Attachment) magnetic disk drives, solid-state drives (SSDs) and other types of storage devices in various combinations.

As noted, processor(s) **105** may also be coupled to a network controller **140** via bus **130**, such as provided by a Network Interface Controller (NIC) that allows the IHS **100** to communicate via an external network, such as the Internet or a LAN. In some embodiments, network controller **140** may be a replaceable expansion card or adapter that is coupled to a motherboard connector of IHS **100**. In some embodiments, network controller **140** may be an integrated component of IHS **100**.

A variety of additional components may be coupled to processor(s) **105** via bus **130**. For instance, processor(s) **105** may also be coupled to a power management unit **165** that may interface with a power supply of IHS **100**. In certain embodiments, a graphics processor **170** may be comprised within one or more video or graphics cards, or an embedded controller, installed as components of the IHS **100**.

In certain embodiments, IHS **100** may operate using a BIOS (Basic Input/Output System) that may be stored in a non-volatile memory accessible by the processor(s) **105**. The BIOS may provide an abstraction layer by which the operating system of the IHS **100** interfaces with the hardware components of the IHS. Upon powering or restarting IHS **100**, processor(s) **105** may utilize BIOS instructions to initialize and test hardware components coupled to the IHS, including both components permanently installed as components of the motherboard of IHS **100** and removable components installed within various expansion slots supported by the IHS **100**. The BIOS instructions may also load an operating system for use by the IHS **100**. In certain embodiments, IHS **100** may utilize Unified Extensible Firmware Interface (UEFI) in addition to or instead of a BIOS. In certain embodiments, the functions provided by a BIOS may be implemented, in full or in part, by the remote access controller **160**. In some embodiments, BIOS may be configured to identify hardware components that are detected as being currently installed in IHS **100**. In such instances, the BIOS may support queries that provide the described unique identifiers that have been associated with each of these detected hardware components by their respective manufacturers. In providing an abstraction layer by which hardware of IHS **100** is accessed by an operating system, BIOS may identify the I/O ports **150** that are recognized and available for use.

In some embodiments, IHS **100** may include a TPM (Trusted Platform Module) that may include various registers, such as platform configuration registers, and a secure storage, such as an NVRAM (Non-Volatile Random-Access Memory). The TPM may also include a cryptographic processor that supports various cryptographic capabilities. In IHS embodiments that include a TPM, a pre-boot process implemented by the TPM may utilize its cryptographic capabilities to calculate hash values that are based on software and/or firmware instructions utilized by certain core components of IHS, such as the BIOS and boot loader of IHS **100**. These calculated hash values may then be compared against reference hash values that were previously stored in a secure non-volatile memory of the IHS, such as during factory provisioning of IHS **100**. In this manner, a TPM may establish a root of trust that includes core components of IHS **100** that are validated as operating using instructions that originate from a trusted source.

In various embodiments, an IHS **100** does not include each of the components shown in FIG. **1**. In various embodiments, an IHS **100** may include various additional components in addition to those that are shown in FIG. **1**. Furthermore, some components that are represented as separate components in FIG. **1** may in certain embodiments instead be integrated with other components. For example, in certain embodiments, all or a portion of the functionality provided by the illustrated components may instead be provided by components integrated into the one or more processor(s) **105** as a systems-on-a-chip (SoC).

A person of ordinary skill in the art will appreciate that IHS **100** is merely illustrative and is not intended to limit the scope of the disclosure described herein. In particular, any computer system and/or device may include any combination of hardware or software capable of performing certain operations described herein. In addition, the operations performed by the illustrated components may, in some embodiments, be performed by fewer components or distributed across additional components. Similarly, in other embodiments, the operations of some of the illustrated components may not be performed and/or other additional operations may be available.

A person of ordinary skill will recognize that IHS **100** of FIG. **1**, which is generally depicted as a server IHS, is only one example of a system in which the certain embodiments of the present systems and methods may be utilized. Indeed, the embodiments described herein may be used in various IHSs, such as a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), or other electronic devices, such as network router devices, televisions, custom telecommunications equipment for special purpose use, etc. That is, certain techniques described herein are in no way limited to use with the IHS of FIG. **1**.

As noted, data, from source to destination, travels through many segments in a data and/or network path. Embodiments of the present systems and methods for end-to-end efficient encryption with security chaining protect data in every data path segment and provides cleartext accessibility for data processing efficiency at data nodes. To wit, embodiments of the present systems and methods provide data end-to-end efficient encryption (E2EEE) and security chaining from a data source (application) system to many compute and/or storage systems in the data and/or network path chain. Embodiments of the present systems and methods are applicable to compute systems with external storages storage replication, edge computing infrastructure with data gathering and/or collection path to data clouds, and/or the like.

FIGS. **2**A and **2**B are partial views intended to form one complete view of a block diagram of IHS network environment **200** showing end-to-end efficient encryption with security chaining, for example (e.g., of a single (serial) segment chain **202**), according to some embodiments. Implementation of embodiments of the present systems and methods result in segments **204** of E2EEE on the interfaces of each IHS **206**, **208***a* through **208***n* which perform metadata processing, handshake signaling, and execution of encryption configuration state machines. Segments **204** of E2EEE on the interfaces of each IHS also perform data block encryption and decryption functions. Embodiments of the present systems and methods also include a security chaining component **210** on every IHS that relays the E2EEE segment between the left-bound and right-bound interfaces, in the respective IHS. On a data-source interface **214** (such as a compute host), embodiments of the present systems and methods may include a metadata and handshake signals generation component that creates and updates metadata and handshake signals and a security operation component that initiates and controls security operations, such as encryption enablement and disablement, rekeying, etc.

E2EEE in accordance with embodiments of the present systems and methods includes a number of data connection segments **204**. In a relative manner, each segment includes left-bound (up-stream) interface **214** and right-bound (down-stream) interface **216**. Thus, data-source IHS **206** (such as a compute host) may host such left-bound (up-stream) interface **206**. Other IHSs in chain **202** provide both left bound and right bound interfaces. Both left bound and right bound interfaces write-encrypt and read-decrypt actions (i.e., functions) in a synchronous manner. The synchronization between left bound and right bound interfaces **214** and **216** is achieved by the combination of in-band (on device or data path) metadata key slot I and II **218** and **220**, out-of-band (control channel) handshake signals **222** and state machine **224**.

Security chaining logic block **210** performs adaptation from the up-stream segment to the down-stream segment. In-band metadata content **226** is immutable (cannot be changed), however, the location of metadata on device (e.g., metadata key slot I **218** or metadata key slot II **220**) is mutable (flexible to change). Handshake signals **222** are regenerated for each segment. Signals **222** for up-stream segment and signals for down-stream segment are synchronized in each IHS. Each segment has a state machine (**224**). State machine **224** uses metadata **226** and signals **222** as input and guides (i.e., directs) write-encrypt and read-decrypt actions for the segment. The state machine for up-stream segment and state machine for down-stream segment are coordinated in each IHS. One up-stream segment can chain and/or bind multiple down-stream segments for data redundancy (i.e., replicate data, **228**). Metadata **226** may be referred to herein as "immutable," meaning metadata **226** is not altered by any down-stream segment (interface) but is propagated. The data originator, data-source IHS **206**, does modify the metadata, so for it, metadata is not "immutable."

Security operations include enabling and disabling of encryption of data, shallow (Key Encryption Key (KEK)) and deep (Data Encryption Key (DEK)) rekeying, etc., for all segments in the data path, as described in greater detail below. Embodiments of the present systems and methods have the following characteristics for all security operations. Any operation, from initiation to completion, involves only processing of control information. The control information includes metadata **226**, handshake signals **222** for every E2EEE segment in the data path, and state machines **224** of both endpoints **214**, **216** for every E2EEE segment **204**. Since only control information processed for every operation, any operation can be completed instantly or in a very short time. Any operation is agnostic to data volume size.

Descriptions herein list some security operations using Small Computer System Interface Logic Unit Number (SCSI LUN) as the example of data volume(s) and SCSI commands as (the) out-of-band control channel(s).

Embodiments of the present systems and methods may (also) support multiple segment chains with single data source IHS. FIGS. **3**A and **3**B are partial views intended to form one complete view of a block diagram of IHS network environment **300** showing multiple parallel end-to-end efficient encryption with security chaining (e.g., of multiple (serial) segment chains **302***a-n*), according to some embodiments.

In accordance with the foregoing, metadata **226** and handshake signals **222** are handled as follows. With respect to in-band metadata **226**, data-source IHS **206** reads and writes metadata **226**. Any other compute or storage IHS down the data path chain (e.g., **208***a* through **208***n*) only reads metadata **226**. In other words, metadata **226** is created and updated by the data source IHS **206** only. In accordance with embodiments of the present systems and methods, E2EEE metadata **226** should, by way of example, include the following.

Volume metadata describes the protected data object and should, by way of example, include: an encryption identifier ("magic" or signature of the metadata); a volume universally unique identifier (UUID) (e.g., a 128-bit label), encryption type and/or capability, an encryption mode; etc.; and reserved space or alignment padding (to isolate volume metadata and encryption metadata I/O blocks, etc.).

Encryption metadata should, by way of example, include: E2EEE state machine input; encryption parameters (2 copies for rekey and state machine transitions), including KEK properties: UUID, algorithm and control info, (optional) Intermedium Key (IK (or MK)) for protection (KEK could be used as IK in some embodiments) and encrypted DEK cryptographically processed control information; (optional) data-source private (e.g., data-source system internal use of different encryption engines); (optional) other systems private (e.g., a replication handshake, etc.); and reserved space and padding.

Metadata protection should, by way of example, include: integrity control with authenticated checksum (e.g., a Hash-Based Message Authentication Code (HMAC), or the like); version control (modification time, sequence number etc.); and E2EEE segment configuration state control with designated data block content verification, or the like.

Metadata Area in the Data Volume:

Size	Name	Description
4K	Metadata Header	See a table below
4K	Crypt Header Slot 1	See a table below
4K	Crypt Header Slot 2	See a table below
128K	Workspace	Used by data-source IHS for reencryption, crypt header editing etc.
TMS-128K-128K-16K	Reserved	Not used in this version
16K	Secondary GPT Table	Located at the last 16K of the device

* TMS—total metadata size

Metadata Header:

Size in bytes	Name	Description
uint8[8]	Header Magic	Also identifies a version of the header format
uint8[16]	Volume ID	Volume GUID
uint16	Crypt Header Type	1 - Linux host, 2 - Windows host
uint16	Crypt Header Slot	Active slot number: 0 - None (initial state, no accepted crypt header in either slot) 1 - Slot 1 (crypt header in slot 1 is actively used) 2 - Slot 2 (crypt header in slot 2 is actively used)
uint16	Crypt Header Size	Actual header size in the active slot in bytes, not including the zero padding
uint8[4]	Crypt Header 1 Checksum	CRC32 sum of the Crypt Header in slot 1.
uint8[4]	Crypt Header 2 Checksum	CRC32 sum of the Crypt Header in slot 2.
uint64	Data Offset (0 based)	Offset of the encrypted data area, in 512-byte blocks starting from the beginning of the device
uint64	Data Size	Size of the encrypted data area, in 512-byte blocks

-continued

Size in bytes	Name	Description
uint16	State	Used by the data-source IHS to persist intermediate states of the volume initialization. This allows the initialization to be resumed if it was interrupted. 0 - Uninitialized (initial state, no crypt header in either slot) 1 - Ready (volume is fully initialized; metadata is consistent and not in transition) 2 - Use Slot 1 (crypt header candidate written to slot 1) 3 - Use Slot 2 (crypt header candidate written to slot 2)
uint8[32]	Authenticated Checksum (covers Metadata Header from offset 0 to the beginning of the Authenticated Checksum)	hmac-sha256 of the metadata using the active Crypt Header Slot KEK
uint8[ ]	Reserved/Padding	Zero Padding to 4096 bytes

Note:
Extra device data, e.g., Source device WWN, Device capability, Array supported encryption algorithms, etc. may be added to the metadata in place of zero padding.

Crypt Header:

Size	Name	Description
uint16	Data encryption algorithm (encrypted by DEK)	"0" = AES-XTS-256 (default) "others" TBD
uint8[16]	KEK ID	KEK Identifier
uint8[12]	IK Nonce	Nonce used for AES-CCM encryption of the IK material
uint8[16]	IK MAC	MAC of the IK material
uint8[32]	IK Data	IK material, AES-CCM encrypted with KEK
uint8[12]	DEK Nonce	Nonce used for AES-CCM encryption of DEK material
uint8[16]	DEK MAC	MAC of DEK material
uint8[64]	DEK Data	DEK material, AES-CCM encrypted with IK
uint8[ ]	Windows host specific key data	Windows only, zeroes for other types of hosts
uint8[ ]	Reserved/Padding	Zero Padding to 4096 bytes

Encryption Keys Specifications:

Key Name	Key Size (bits)	Protection
KEK (Key Encryption Key)	256	Key store protected
IK	256	AES-CCM encrypted with KEK
DEK (Data Encryption Key)	512	AES-CCM encrypted with IK

Metadata protection may be provided by an IK for metadata protection and in accordance with embodiments of the present systems and methods, metadata has integrity checksum, with integrity of the checksum. Also, in accordance with embodiments of the present systems and methods, critical metadata is protected. Metadata protection is also provided by the IK being internal to metadata **226** creation and validation (e.g., in data-source IHS host **206**).

Volume UUID generation and regeneration may include the data-source (e.g., in data-source IHS host **206**) generating the volume UUID and saving it in metadata **226** when the volume transits from the unencrypted to encrypted state, as discussed below. The data-source (**206**) may (also) optionally regenerate the volume UUID and save it and old UUID(s) on metadata **226**, such as when (or if) the volume is a mounted snapshot, a linked volume, a replica (e.g., **228**), or the like. Thereby, metadata **226**, as implemented in accordance with embodiments of the present systems and methods, provides a mechanism to distinguish the original data volume from a mounted snapshot, linked volume, replica (**228**) or the like. For example, data-source IHS **206** may compare a device physical ID saved in metadata **226** with current device physical ID as currently reported for the data in question and may consider a mismatch as a cloned volume case, or the like. When data-source IHS **206** regenerates the volume UUID, the data-source IHS also updates and/or adds the new native device physical ID in metadata. Thusly, the volume becomes a new source (original) volume.

For handshake signal over control channel implementation, each handshake signal **222** carries the following state info:

Parameter	Description	Default Value	Initiator Access	Target Access
Use Slot	Set by initiator to 1 or 2 to indicate the Crypt Header Slot that should be used by the target. It represents a desired configuration.	0	read/write	read
Active Slot	Crypt Header Slot number (1 or 2) that is currently used by the target. It represents the actual configuration. The value 0 indicates no active configuration.	0	read	write
Config Error	Set by the target to an error code if target failed to reach the desired configuration which is set in Use Slot. Set to 0 before starting a configuration process.	0	read	write

Error Code Descriptions:

Error Code	Description
0	No error
01h	Generic error
02h	Metadata read/decode error
03h	KEK request error
04h	DEK unwrap error
05h	Invalid Data Offset or Data Size

A handshake (signals **222**) implementation example, using SCSI commands may: use "INQUIRY" command to retrieve generic volume parameters, such as capacity; use "MODE_SELECT" to set volume parameters, such as on a vendor-specific page; use "MODE_SENSE" to read volume parameters, such as on a vendor-specific page; and/or uses "Mode Vendor Specific: page "20h," as follows:

	Bit							
Byte	7	6	5	4	3	2	1	0
0	PS (0)	Page Code (20 h)						
1	Parameter List Length (02 h)							
2	Reserved				Active Slot		Use Slot	
3	Config Error							

FIG. **4** is a block diagram of example end-to-end efficient encryption segment **204**, showing functions therein, according to some embodiments. With respect to communications for encryption control in accordance with embodiments of the present systems and methods, in-band metadata **226** is persistent on the data volume (e.g., SCSI LUN volume, Non-Volatile Memory express (NVMe) volume, or the like). Out-of-band (or out-of-data) handshake signals **222** are encryption control signals carried on networking protocols, such as SCSI/iSCSI commands, NVMe commands, Transmission Control Protocol/Internet Protocol (TCP/IP) connection, or the like. Handshake signals **222** are also "out-of-data" in that they are not part of the (encrypted) data volume, unlike metadata **226**, which is generated in a trailer of an (encrypted) data volume.

As shown in FIG. **4**, tasks performed on upstream, left-bound interface **214** may include: intercept **402** the write to data volume **404** and encrypt the data; intercept the read and decrypt data; metadata processing **406**, which includes building and updating encryption metadata **226** at the end of data volume **404** that defines the encryption of volume **404** (e.g. slot I **218** or slot II **220**); exchange **408** handshake signals **222** with the connected storage for encryption enablement or disablement, rekey, and other security operations and perform security operations for data volume(s) **404**. Such security operations for the data volume(s) may include creation of a new empty volume, maintenance of an existing encrypted volume mounted to the host IHS; existing data encryption and/or decryption, such as, data transition from unencrypted to encrypted and from encrypted to unencrypted; shallow rekeying (as described below); or deep rekeying (as described below). Tasks performed on downstream, right-bound interface **216** may include intercept **410** the write from the host (**206**) and decrypt data **404**; intercept **410** the read from the host (**206**) and encrypt data **404**; metadata processing **406** to detect and read metadata **226** at the end of the data volume **404** that identify the encryption (e.g. slot I **218** or slot II **220**); exchange **408** handshake signals **222** for encryption enablement or disablement, rekey, and other security operations (such as described below).

FIG. **5** is a block diagram of the end-to-end efficient encryption segment of FIG. **4**, showing communications within example segment **204**, according to some embodiments. Therein, communication steps **504** through **514**, are described below, between the host IHS (upstream, left-bound interface **214**) and storage (downstream, right-bound interface **216** are shown), at a high level. Key Management System (KMS) **502** may be external to IHSs **206** through **208***n* in segment chain **202**, as shown. KMS **502** generates and stores KEKs. KMS **204**, may, in accordance with various embodiments of the present systems and methods, KMS be a process that is hosted by an IHS, or the like and may be full-time network-accessible to each E2EEE segment **204**. For example, KMS **502** may be a Key Management Interface Protocol (KMIP) server, a cloud based Key Vault or Manager, or the like, and each segment **204** may communicate with the KMS to obtain the KEK. For example, in accordance with embodiments of the present systems and methods, upstream, left-bound interface **214**, requests (**504**) the KEK from KMS **502**. Upstream, left-bound interface **214** then initiates security actions and communications with storage, by creating and updating encryption metadata **226** and handshake indications in the metadata (**506**) and sending (**508**) handshake signals via out-of-band communications **510**. Right-bound (downstream) interface **216** retrieves (**512**) the KEK from KMS, unwraps the DEK and performs security actions by following encryption metadata **226** and handshake signals **222**, thereby the current storage (i.e., down-stream, right-bound interface **216**) may perform security chaining communications with next storage interface. Upstream, left-bound interface **214** then queries (**514**) right-bound (down-stream) interface **216** concerning states and/or status, via out-of-band communications **510** to complete security operations, such as enablement or disablement, rekey, etc.

FIG. **6** is a block workflow diagram of end-to-end efficient encryption security chaining **600**, according to some embodiments. In accordance with embodiments of the present systems and methods, security chaining **600** takes place within the same IHS, and is the segment relay between right-bound interface communications to left-bound interface communications and between left-bound interface communications to right-bound interface communications.

In accordance with E2EEE security chaining under embodiments of the present systems and methods, in-band metadata **226** is not changed other than by the originator (e.g., data-source IHS **206**). Such metadata changes by the originator are inherently, in accordance with embodiments of the present systems and methods, propagated down the chain. Also, in accordance with E2EEE security chaining under embodiments of the present systems and methods, every IHS within the same security chain is responsible for performing handshakes (i.e., handshake signaling **222**) with its right-bound neighbor and for suspending downstream I/O until the handshake is completed. Handshake signals **222** trigger state machine (**224**) transitions. As noted, handshake signals **222** are communicated between right and left-bound interfaces (IHSs) of each E2EEE segment via out-of-band (out-of-data) control channel, such as SCSI commands, NVMe commands, and TCP/IP connection messages.

Also, under security chaining **600**, all security operations (e.g., enable or disable encryption, shallow and/or deep rekey, etc., described below) follow the same handshake procedures across chain **202**. Thereunder, security operation **602** is initiated by data-source IHS **206**, as depicted in the FIG. **6**. Node 1 (data-source IHS **206**) and node 2 (IHS **208***a*) synchronize the handshake signals (**222**) between 1r **604** (right endpoint (of a left-bound (up-stream) interface **214**)) to 2l **606** (left endpoint (of a right-bound (down-stream) interface **216**)). Node 2 (IHS **208***a*) and node 3 (IHS **208***b*) synchronize the handshake signals (**222**) between 2r **608** (right endpoint (of a left-bound (up-stream) interface **214**)) to 3l **610** (left endpoint (of a right-bound (down-stream) interface **216**)). Node 3 (IHS **208***b*) and node 4 (not shown (IHS **208***c*)) synchronize the handshake signals (**222**) between 3r **612** (right endpoint (of a left-bound (up-stream) interface **214**)) to 4l (not shown, left endpoint (of a right-bound (down-stream) interface **216**)). This continues until the last E2EEE segment in the chain (**202**).

Chaining logic block **614** in each node handles the handshake (**222**) changing synchronization between the left E2EEE segment and the right E2EEE segment on each node (i.e., between 2l **606** and 2r **608**, and between 3l **610** and 3l **612**). The keys, KEK provided via KMS **502**, and the DEK derived therefrom and stored in the metadata (**226**), are used for the encryption and decryption across the chain (**202**) and are each the same. All security operations under embodiments of the present systems and methods are instant or fast, since only the metadata (**226**), handshake signals (**222**), and state machines (**224**) are synchronized, there is no need to encrypt and/or decrypt the actual storage data, since the operations only apply to the I/O path between each segment.

State machine(s) **224** is (are) discussed in greater detail below with respect to enablement and disablement of encryption, and shallow and/or deep rekeying. However, FIG. **7** is a (complete) compute host and storage IHS state machine diagram for end-to-end efficient encryption with security chaining, according to some embodiments, showing states **702** through **722** discussed below. The handshake process is based on state machine **224**, for which, the inputs are the in-bound metadata (**226**) and the handshake signals (**222**) from the left-bound (up-stream) interface **214**, wherein the states reflect the left-bound E2EEE segment encryption configuration. Security chaining **600** provides data path High Availability (HA) for dynamically connecting multiple right-bound IHSs **208***a-n*. The data blocks are replicated to all downstream IHSs **208***a-n* in accordance with embodiments of the present systems and methods. This enables replacement of an IHS, transparently, and/or enables implementation of data path redundancy (**228**).

Security operations in embodiments of the present systems and methods include enabling and disabling of encryption of data, shallow (KEK) and deep (DEK) rekeying, etc., for all segments (**204**) in the data path (**202**) Embodiments of the present systems and methods have the following characteristics for all security operations. Any operation, from initiation to completion, involves only processing of control information. The control information includes metadata **226**, handshake signals **222** for every E2EEE segment **204** in data path **202**, and state machines **224** of both endpoints for every E2EEE segment **204**. Since only control information is processed for every operation, any operation can be completed instantly or in a very short time. Also, in accordance with embodiments of the present systems and methods, any operation is agnostic to data volume size, and the like.

FIGS. **8**A and **8**B are partial views intended to form one complete view of a block diagram of an IHS network environment, such as of single (serial) segment chain **202** of FIGS. **2**A and B, showing enablement **800** of end-to-end efficient encryption with security chaining, according to some embodiments. Therein data source (host) **206** will write metadata **226** containing and employing key slot I **218**, or the like, in a trailer of a data block using the key slot. Security chaining **220**, as discussed above, ensures metadata **226** remains unmutated (unchanged) to (the) down-stream segment(s). State machine **224** is set at an "initial state," herein US (use slot)=0, AS (active slot)=0 (**802**). FIG. **9** is a state machine diagram for enablement of end-to-end efficient encryption with security chaining, according to some embodiments, such as states **702** though **708** of FIG. **7**, wherein 702 shows initial state US=0, AS=0. No actual encryption or decryption occurs in this regard.

Data Source (host) **206** then sends signal **222** (**804**) "Use slot I," or the like. Security chaining **220** regenerates signal "Use slot I" for next segment. State machine changes (**806**) from US=0, AS=0 (**702**) through US=1, AS=0 (**704**) to US=1, AS=1 (**706**). Security chaining, as discussed above, ensures this change occurs (backward) from (the last) segment to (the first) segment. Again, no actual encryption or decryption occurs in this regard.

Data source (host) **206** then starts writing/reading user data, all IHSs in the chain encrypt/decrypt user data (e.g., decrypt user data when accessed (read) and encrypts user data when added or changed (written)) otherwise data is sent between the segments in encrypted format. Thusly, security chaining, as discussed above, enables encryption and decryption (forward) from (the first) segment to (the last) segment, and encryption/decryption applies to all data writing and reading in the chain (**202**).

FIG. **10** is a swim lane workflow diagram showing enablement **1000** of end-to-end efficient encryption with security chaining, such as of single (serial) segment chain **202** of FIGS. **2**A and B, according to some embodiments E2EEE enablement **1000** is the procedure to enable volume

EE (online transition from unencrypted to EE encrypted). The description below assumes the data volume to be encrypted has enough space for metadata **226** at the volume trailer. Enablement procedure **1000** covers both new and/or empty volume encryption and existing unencrypted data-set volume encryption.

Upon a request to enable encryption at **1002**, an exchange of information with KMS for key (KEK) creation and distribution takes place. To identify the key (KEK), data-source IHS **206** requests a key from the KMS at **1004** and each of the IHSs in chain **202** retrieves the key. At **1006**, data-source IHS **206** creates encryption metadata **226** at the trailer of the volume for which encryption is being enabled. For existing unencrypted data-sets volume encryption, if there is not enough empty space for metadata at the end of the volume, data-source IHS **206** may, in accordance with various embodiments of the present systems and methods, either shrink the volume or request storage to add extra space. The right-bound traffic should be suspended, at **1008**, to prevent any (new) data to be transmitted insecurely downstream. The encrypted volume (OS-native) metadata and volume format are prepared. However, all metadata is generated at **1006** before enabling actual encryption, the actual data encryption is enabled independently from metadata creation and control, Also the encrypted volume is provided access to applications.

Handshakes for all E2EEE segments, beginning with handshake **1010**, are completed. Handshakes (signals **222**) are carried out via out-of-band (out-of-data) communications, (such as trough SCSI commands), as noted. This enablement procedure can be completed "instantly" (i.e., over a short period of handshaking time). There is no need to encrypt, or otherwise "walk-through" any existing data-sets.

Security chaining logic **220** is responsible for replication of encryption metadata **226** to the right endpoint(s) (left-bound interface(s) **214**) as well as generating appropriate control signals **222** to coordinate reconfiguration of the right encryption endpoint with the left-bound encryption endpoint (right-bound interface **216**) of the downstream IHS(s). Security chaining logic **220** will not propagate metadata changes until the left endpoint (right-bound interface **216**) successfully finishes the handshake. That is, until both sides confirm re-configuration **1012**, metadata state is "ready" at **1014**. When the metadata is ready at **1014**, chaining logic **220** makes a snapshot of the metadata at **1016** to memory and presents it to the right-bound endpoint (left-bound interface **214**) as encryption metadata **226** at **1018**. It then initiates right-bound handshake **1020**, which is minimal, in accordance with the foregoing, since metadata **226** is in the ready state and does not need multistage updates accompanied by related control signals, or the like, as is typical.

In the case when the right-bound endpoint (left-bound interface **214**) receives a metadata update **1018**, chaining logic **220** should interrupt and roll back the right-bound handshake **1020** if there is one in progress and wait for the ready state metadata **1022**, **1024** on the left-bound endpoint (right-bound interface **216**). As noted, the right-bound traffic should be suspended **1008**, before left-bound traffic is resumed at **1026** through **1030**, to prevent any new data to be transmitted insecurely downstream until after compute host **206** has received a confirmation of the secure channel (i.e., ready **1032**, via acknowledgement **1034**).

The resulting encryption enablement in accordance with embodiments of the present systems and methods are compatible with IHS operating system full volume and block encryption features, supports empty volumes or volumes with data, data appears (i.e., is) encrypted (relatively) instantly, cloning and snapshots are supported, the same encryption function is employed in each segment and chain segments are replaceable.

FIGS. **11**A and **11**B are partial views intended to form one complete view of a block diagram of an IHS network environment, such as of single (serial) segment chain **202** of FIGS. **2**A and B, showing disablement **1100** of end-to-end efficient encryption with security chaining, according to some embodiments. FIG. **12** is a state machine diagram for disablement of end-to-end efficient encryption with security chaining, such as states **702**, **708** and **710** of FIG. **7**, wherein, **708** shows state machine **224** as set at a(n) (initial) state where US (use slot)=1, AS (active slot)=1, (**1102**) according to some embodiments. FIG. **13** is another state machine diagram for disablement of end-to-end efficient encryption with security chaining, such as states **702**, **712** and **714** of FIG. **7**, wherein, **712** shows state machine **224** as set at a(n) (initial) state where US=2, AS=2, according to some embodiments.

To begin disablement **1100** data source (host) **206** sends signal (**222**) "Use slot 0," at **1104**. Security chaining **220** regenerates signal "Use slot 0" for the next segment (**204**), such that state machine **224** changes state from US=1, AS=1 (or US=2, AS=2, etc.) to US=0, AS=0 (at **1106**). Security chaining **220** ensures this change is applied (backward) from (the last) segment to (the first) segment and as a result, encryption and/or decryption is no longer applied to data I/O. Then data Source (host) **206** removes metadata **226** and all data read by host **206** appears (i.e., is) unencrypted "instantly" (i.e., over a short period of handshaking time).

FIG. **14** is a swim lane workflow diagram showing disablement **1400** of end-to-end efficient encryption with security chaining, such as of single (serial) segment chain **202** of FIGS. **2**A and B, according to some embodiments. E2EEE disablement **1400** is the procedure to disable EE (online transition from EE to unencrypted). Encryption disablement procedure **1400** decrypts the encrypted volume in-place. Upon receipt of instruction **1402** to disable end-to-end efficient encryption, compute host **206** initiates update to metadata **226**. The right-bound traffic should be suspended, at **1404**, to prevent any (new) (encrypted) data to be transmitted downstream. Handshakes **1406** setting "Use slot 0" (US=0, AS=0 state) for all E2EEE segments **204** are completed), which results in I/O suspension **1404** and deinitialization of encryption **1408**. Again, the handshakes (signals) **222** are carried out via out-of-band (out-of-data) communications, (such as SCSI commands Then sanitization (clearing) **1410** of encryption metadata is carried out, as described above. As noted, this disablement procedure can be completed "instantly" (over a short period of handshaking time) and there is no need to decrypt and/or "walk-through" any existing datasets.

Security chaining logic **220** is responsible for clearing the encryption metadata area presented to the right-bound endpoint ((left-bound interface **214**), such as through reply **1412**, as well as generating appropriate control signals **222** to coordinate reconfiguration of the right-bound encryption endpoint (left-bound interface **214**) with the left-bound encryption endpoint (right-bound interface **216**) of the downstream IHS(s). Security chaining logic will not clear the encryption configuration of the right-bound segment until the left-bound endpoint (right-bound interface **216**) successfully disables encryption at **1408**, such as by both sides confirming re-configuration metadata is empty or removed, via ready **1414**). When the metadata is emptied or removed, the chaining logic presents empty metadata to the right-bound endpoint (left-bound interface **214**) and initiates a (minimal) handshake **1416**, since no metadata changes are required, and no keys requested.

In the case when a right-bound endpoint (left-bound interface **214**) starts a new handshake (such as to enable encryption again), the chaining logic interrupts any right-bound handshake in progress, keeping I/O suspended, and wait for the ready state metadata on the left-bound endpoint (right-bound interface **216**), at **1414**, and then start a new handshake **1416**. As noted, the right-bound traffic should be suspended **1404**, before left-bound traffic is resumed at **1418** through **1422**, to prevent any new data being encrypted (downstream until) after compute host **206** has received a confirmation of disablement of encryption (i.e., ready **1424**, via acknowledgement **1426**).

Thus, in accordance with embodiments for disablement of end-to-end efficient encryption with security chaining in accordance with embodiments of the present systems and methods, data appears (i.e., is) decrypted instantly, the same function is used in each segment **204** and chain **202** segments **204** remain replaceable.

FIGS. **15**A and **15**B are partial views intended to form one complete view of a block diagram of an IHS network environment, such as of single (serial) segment chain **202** of FIGS. **2**A and B, showing rekeying **1500** in end-to-end efficient encryption with security chaining, according to some embodiments. In accordance with embodiments of the present systems and methods, rekey **1500** (shallow or deep) occurs "instantly" (i.e., over a short period of handshaking time) between all left and right endpoint pairs in the security chain. Any further data manipulation is not required. Rekeying in accordance with embodiments of the present systems and methods, may be used to provide "key rotation," changing the encryption key used to encrypt and decrypt data. Shallow rekey rekeys the KEK and any intermediate key(s), and deep rekey rekeys the KEK, IKs, and DEK. FIG. **16** is a state machine diagram for rekeying in end-to-end efficient encryption with security chaining, such as states **708**, **712**, **716** and **718** of FIG. **7**, wherein, **708** shows state machine **224** as set at a(n) (initial) state where US (use slot)=1, AS (active slot)=1, according to some embodiments. FIG. **17** is another state machine diagram for rekeying in end-to-end efficient encryption with security chaining, such as states **708**, **712**, **720** and **722** of FIG. **7**, wherein, **712** shows state machine **224** as set at a(n) (initial) state where US (use slot)=2, AS (active slot)=2, according to some embodiments.

Initial conditions **1502**, in the example of FIGS. **15**A and B, are shown as key slot I **218** is used and active (i.e. (use slot)=1, AS (active slot)=1 (as in FIG. **16**, at **708**), such that encryption and/or decryption is occurring on all data in chain **202** with the keys (KEK, IKs, and DEK) in key slot I **218**. To rekey, the volume is locked (no reading and/or writing by application(s)) and data source (host) **206** writes new metadata, at **1504** employing metadata key slot II **220**. Security chaining logic block **220** ensures new metadata **226** propagates across chain **202** unmutated (same in all segments **204** across chain **202**). State machine **224** remains at initial state (e.g., US=1, AS=1) **708** and encryption and/or decryption continues occurring on all data in the chain employing key slot I **218**. However, data source **206** then sends signal **222** "Use slot from I to II," **1506** or the like, and security chaining **220** regenerates signal "Use slot from I to II" **1506** for next segment **204**. State machine **224**, in response, changes state from US=1, AS=1 (**1502**) to transit state US=2, AS=1, see, by way of example block **716** of FIGS. **7** and **16**, then to US=2, AS=2 **712**. Security Chaining ensures this change (**1508**) occurs (backward) from (the last) segment to (the first) Segment. Thereby, (the last) segment to (the first) segment progressively switch to key slot II **220**. Data source **206** then unlocks the volume and starts writing and/or reading user data, and all IHSs in chain **202** encrypt and/or decrypt user data, as called for using the new key(s) (e.g., decrypt user data when accessed (read) and encrypts user data when added or changed (written)) otherwise data is sent between the segments in encrypted format. Thusly, security chaining enables encryption and/or decryption (forward) from (the first) segment to (the last) segment, and encryption and/or decryption occurs in chain **202** using keys in key slot II **220**.

As noted, efficient encryption rekeying in accordance with embodiments of the present systems and methods may support both shallow rekey (KEK) and deep rekey (KEK and DEK) operations. In accordance with various embodiments, rekey of intermedium key (IK) (if used) may be included in either shallow rekey or deep rekey. Embodiments of the present systems and methods may, support two sets of keys for "Atomic" metadata updates (i.e., indivisible and irreducible (series of) updates such that either all metadata updates occur, or no metadata update occurs), in the manner described above, one active key and one unused key, or one active key and one transition key.

FIG. **18** is a swim lane workflow diagram showing rekeying **1800** in end-to-end efficient encryption with security chaining, such as of single (serial) segment chain **202** of FIGS. **2**A and B, according to some embodiments. Both shallow rekeying (to rekey the KEK) and deep rekeying (to rekey the KEK and DEK) employ procedure **1800**. Therein, upon shallow or deep rekey request **1802** compute host **206** requests a key from the KMS at **1804** and each of the IHSs in chain **202** retrieves the key updates **1804**. Compute host **206** updates encryption metadata **226** at the trailer of the volume at **1806**, The right-bound traffic should be suspended, at **1808**, to prevent any (new) mis-encrypted data being transmitted downstream. Handshakes **1810**, **1812** and **1814** are completed for all E2EEE segments. Again, handshakes **1810**, **1812** and **1814** (signals **222**) are via out-of-band (out-of-data) communications, (such as SCSI commands). Rekey (shallow or deep) occurs "instantly" (i.e., over a short period of handshaking time) between all left and right endpoint pairs in the security chain. Any further data manipulation is not required. For example, there is no need to re-encrypt and/or "walk-through" any existing datasets.

Security chaining logic **220** is responsible for replication of encryption metadata **226** to the right endpoint(s) (left-bound interface(s) **214**) as well as generating appropriate control signals **222** to coordinate reconfiguration of the right encryption endpoint with the left-bound encryption endpoint (right-bound interface **216**) of the downstream IHS(s). Security chaining logic **220** will not propagate metadata changes until the left endpoint (right-bound interface **216**) successfully finishes the handshake. That is, until both sides confirm re-configuration **1816**, metadata state is "ready" at **1818**. When the metadata is ready at **1818**, security chaining logic **220** copies the metadata at **1820** and replays **1812** the rekey handshake whether it be shallow or deep from left-bound endpoint to right-bound endpoint. In accordance with various embodiments of the present systems and methods, this replay of the rekey handshake may be asynchronously since the replication, itself, may be asynchronous. Chaining logic **220** then initiates right-bound handshake **1814**, which is minimal, in accordance with the foregoing, since metadata **226** is in the ready state and does not need multistage updates accompanied by related control signals, or the like, as is typical. However, metadata updates which already contain the new key (KEK, IK, DEK, or all of them) at the beginning of this procedure are used by the chaining logic **220** (unmodified) across the chain between right-bound endpoints and left-bound endpoints.

In the case when the right-bound endpoint (left-bound interface **214**) receives a metadata update **1812**, chaining logic **220** should interrupt and roll back the right-bound handshake **1814** if there is one in progress and wait for the ready state metadata **1822**, **1824** on the left-bound endpoint (right-bound interface **216**). As noted, the right-bound traffic should be suspended **1808**, before left-bound traffic is resumed at **1826** through **1830**, to prevent any (new) mis-encrypted data being transmitted downstream, until after compute host **206** has received a confirmation of the secure channel (i.e., ready **1832**, via acknowledgement **1834**).

In accordance with embodiments of the present systems and methods, storage-based backup volumes, such as backup volumes mounted from a snapshot, cloned volumes, volumes mounted from replicas, and the like may be treated, in accordance herewith, as independent volumes from an original or source volume. In-band metadata on the volume may have enough information for embodiments of the present systems and methods to retrieve keys and encryption states to carry out (restart) efficient encryption procedures in accordance with embodiments of the present systems and methods. Resultingly, embodiments of the present systems and methods provide backup volume security, such as for volumes created during the original volume encryption enablement and rekeying and/or for volumes created after the original volume is encrypted and rekeyed.

Advantageously, embodiments of the present systems and methods achieve data confidentiality across the complete data path chain from the data source to final storage destination and still allows data processing efficiency in the IHSs on the data path chain. Embodiments of the present security chaining allows data confidentiality from one IHS to another and still allows data processing efficiency in all IHSs in the middle. Embodiments of the present systems and methods achieves "instant" (i.e., over a short period of handshaking time) data encryption operations (such as encryption enablement and/or disablement and/or shallow or deep rekeying) between compute host and host connected storage system and among chained compute and/or storage IHSs. A separate handshake control channel delivers flexible and robust security operation triggers (such as encryption enablement or disablement, shallow and deep rekey) across the complete data path chain, in embodiments of the present systems and methods. Security chaining in accordance with embodiments of the present systems and methods, allows addition, removal, and replacement of E2EEE segments, dynamically, without the need to reconfigure the entire security chain, as required in typical encryption schemes. The present handshake and state machine implementation is agnostic of any encryption function. The reason for a state transition and the effect of it is beyond the scope of an IHS in the chain. Only the metadata originator possesses this information and implements the higher-level logic such as applying encryption to the data blocks or changing the encryption key.

Embodiments of the present systems and methods may generally be applied to any distributed infrastructure with a data path that includes two or more IHSs. For example, for a compute server and external storage with replication, the compute server may be a Linux® or Windows® based IHS, a Kubernetes application with a Container Storage Interface (CSI) persistent volume, a Hypervisor node, or the like. Storage systems that may employ embodiments of the present systems and methods may include primary storage and remote replication storage. Edge computing infrastructure that includes (an) edge device(s) and/or server, edge gateway, edge data center, and cloud storage may employ embodiments of the present systems and methods.

It should be understood that various operations described herein may be implemented in software executed by processing circuitry, hardware, or a combination thereof. The order in which each operation of a given method is performed may be changed, and various operations may be added, reordered, combined, omitted, modified, etc. It is intended that the invention(s) described herein embrace all such modifications and changes and, accordingly, the above description should be regarded in an illustrative rather than a restrictive sense.

Reference is made herein to "configuring" a device or a device "configured to" perform some operation(s). It should be understood that this may include selecting predefined logic blocks and logically associating them. It may also include programming computer software-based logic of a retrofit control device, wiring discrete hardware components, or a combination of thereof. Such configured devices are physically designed to perform the specified operation(s).

Modules implemented in software for execution by various types of processors may, for instance, include one or more physical or logical blocks of computer instructions, which may, for instance, be organized as an object or procedure. Nevertheless, the executables of an identified module need not be physically located together but may include disparate instructions stored in different locations which, when joined logically together, include the module and achieve the stated purpose for the module. Indeed, a module of executable code may be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices. Similarly, operational data may be identified and illustrated herein within modules and may be embodied in any suitable form and organized within any suitable type of data structure. The operational data may be collected as a single data set or may be distributed over different locations including over different storage devices.

To implement various operations described herein, computer program code (i.e., instructions for carrying out these operations) may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, Smalltalk, Python, C++, or the like, conventional procedural programming languages, such as the "C" programming language or similar programming languages, or any of machine learning software. These program instructions may also be stored in a computer readable storage medium that can direct a computer system, other programmable data processing apparatus, controller, or other device to operate in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the operations specified in the block diagram block or blocks. The program instructions may also be loaded onto a computer, other programmable data processing apparatus, controller, or other device to cause a series of operations to be performed on the computer, or other programmable apparatus or devices, to produce a computer implemented process such that the instructions upon execution provide processes for implementing the operations specified in the block diagram block or blocks.

The terms "tangible" and "non-transitory," as used herein, are intended to describe a computer-readable storage medium (or "memory") excluding propagating electromagnetic signals; but are not intended to otherwise limit the type of physical computer-readable storage device that is encompassed by the phrase computer-readable medium or memory. For instance, the terms "non-transitory computer readable medium" or "tangible memory" are intended to encompass types of storage devices that do not necessarily store information permanently, including, for example, RAM. Program instructions and data stored on a tangible computer-accessible storage medium in non-transitory form may afterwards be transmitted by transmission media or signals such as electrical, electromagnetic, or digital signals, which may be conveyed via a communication medium such as a network and/or a wireless link.

Unless stated otherwise, terms such as "first" and "second" are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements. The terms "coupled" or "operably coupled" are defined as connected, although not necessarily directly, and not necessarily mechanically. The terms "a" and "an" are defined as one or more unless stated otherwise. The terms "comprise" (and any form of comprise, such as "comprises" and "comprising"), "have" (and any form of have, such as "has" and "having"), "include" (and any form of include, such as "includes" and "including") and "contain" (and any form of contain, such as "contains" and "containing") are open-ended linking verbs. As a result, a system, device, or apparatus that "comprises," "has," "includes" or "contains" one or more elements possesses those one or more elements but is not limited to possessing only those one or more elements. Similarly, a method or process that "comprises," "has," "includes" or "contains" one or more operations possesses those one or more operations but is not limited to possessing only those one or more operations.

Although the invention(s) is/are described herein with reference to specific embodiments, various modifications and changes can be made without departing from the scope of the present invention(s), as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present invention(s). Any benefits, advantages, or solutions to problems that are described herein with regard to specific embodiments are not intended to be construed as a critical, required, or essential feature or element of any or all the claims.

The invention claimed is:

1. A method for rekeying an information handling system network end-to-end efficient encryption with security chaining, the method comprising:

locking an encrypted data volume, wherein a currently used and active metadata key slot is in use, thereby preventing reading from, and writing to, the encrypted data volume by applications;

writing, by a data source information handling system, new metadata in a trailer of an encrypted data block of the encrypted data volume using a key slot different than the currently used and active metadata key slot;

sending, out-of-band by the data source information handling system, an out-of-data control signal to change the key slot in use for the encrypted data volume to the key slot different than the currently used and active metadata key slot; and

unlocking, by the data source information handling system, the encrypted data volume, enabling writing and/or reading user data by the data source information handling system, encryption and decryption in all information handling system end-to-end efficient encryption data connection segment interfaces.

2. The method of claim **1**, further comprising requesting, by the data source information handling system, a new key from a key management system and updating the different key slot with the new key.

3. The method of claim **1**, further comprising, maintaining, by a security chaining logic component in each network information handling system, the new metadata across all data connection segment interfaces of the network information handling systems.

4. The method of claim **1**, further comprising, prior to unlocking the encrypted data volume, maintaining a state machine of each information handling system end-to-end efficient encryption data connection segment interface in a state with the currently used and active metadata key slot, encryption and decryption prior to unlocking the encrypted data volume occurring on all data in the network using the currently used key slot.

5. The method of claim **1**, further comprising, regenerating and sending out-of-band, by an information handling system security chaining logic component in each network information handling system, the out-of-data control signal to change the key slot in use from the currently used key slot to the different key slot to each next information handling system end-to-end efficient encryption data connection segment interface.

6. The method of claim **1**, further comprising, changing, in an encryption configuration state machine of each information handling system end-to-end efficient encryption data connection segment interface, in response to the out-of-data control signal to change the key slot in use from the currently used key slot to the different key slot, a state with the currently used and active metadata key slot to use the different metadata key slot and make the different metadata key slot the active key slot.

7. The method of claim **6**, further comprising, transiting, in the encryption configuration state machine of each data connection segment interface, in response to the out-of-data control signal to change the key slot in use from the currently used key slot to the different key slot, a state with the currently used and active metadata key slot to use the different metadata key slot and maintain the currently active metadata key slot, prior to making the different metadata key slot the active key slot, maintaining encryption and decryption prior to unlocking the encrypted data volume occurring on all data in the network using the currently active key slot.

8. The method of claim **1**, further comprising, enabling, by a security chaining logic component in each network information handling system, changing an encryption configuration state machine of each data connection segment interface to a state to use the different metadata key slot and making the different metadata key slot the active key slot, backward from a last data connection segment interface to a first data connection segment interface, such that the last data connection segment interface to the first data connection segment interface progressively switch to the different key slot.

9. The method of claim **1**, further comprising, enabling, by a security chaining logic component in each network information handling system, encryption and decryption in all information handling system end-to-end efficient encryption data connection segment interfaces forward from a first data connection segment interface to a last data connection segment interface, such that encryption and decryption in all information handling system end-to-end efficient encryption data connection segment interfaces use the different key slot.

10. The method of claim **1**, wherein the rekeying is shallow, rekeying a key encryption key and any intermediate key.

11. The method of claim **1**, wherein the rekeying is deep, rekeying a key encryption key, any intermediate key or keys and a data encryption key.

12. The method of claim **1**, further comprising, in response to a first data connection segment interface receiving a metadata update, a security chaining logic component, interrupting and rolling back an out-of-data control signal to change the key slot in use from the currently used key slot to the different key slot to a second data connection segment interface, and waiting for a ready state in the metadata on the second data connection segment interface to unlock the encrypted data volume.

13. A system to rekey end-to-end efficient encryption (E2EEE) with security chaining in an information handling system (IHS) network comprising a plurality of IHSs, the system comprising:

a data source information handling system (DSIHS) comprising:

one or more DSIHS processors; and

a DSIHS non-transitory computer readable medium operably coupled with the one or more DSIHS processors, the DSIHS non-transitory computer readable medium storing processor executable program instructions configured to cause the DSIHS to perform operations when the processor executable program instructions are executed by the one or more DSIHS processors, the operations comprising:

lock an encrypted data volume, wherein a currently used and active metadata key slot is in use, thereby preventing reading from, and writing to, the encrypted data volume by applications;

write new E2EEE metadata in a trailer of an encrypted data block of the encrypted data volume using a different key slot than the currently used and active metadata key slot;

send an out-of-data control signal out-of-band to change the key slot in use from the currently used key slot to the different key slot; and

unlock the encrypted data volume, enabling writing and/or reading user data, encryption and decryption by the DSIHS via a in E2EEE data connection segment interface (DCSI) configured in each IHS of the plurality of IHSs;

a DCSI state machine in each DCSI of a plurality of IHS E2EEE DCSIs configured in the respective plurality of IHSs, each DCSI state machine comprising:

one or more DCSI processors; and

a DCSI state machine non-transitory computer readable medium operably coupled with the one or more DCSI processors, the DCSI state machine non-transitory computer readable medium storing processor executable program instructions configured to cause the DCSI state machine to perform operations when the processor executable program instructions are executed by the one or more DCSI processors, the operations comprising:

receive an out-of-data control signal out-of-band, wherein the out-of-data control signal is configured to cause the DCSI state machine to change the key slot in use from the currently used key slot to the different key slot; and

in response to the out-of-data control signal, change a state of the DCSI state machine to use the different metadata key slot and make the different metadata key slot the active key slot; and

a security chaining logic component (SCLC) in each IHS of the plurality of IHSs, each SCLC comprising:

one or more SCLC processors; and

a SCLC non-transitory computer readable medium operably coupled with the one or more SCLC processors, the SCLC non-transitory computer readable medium storing processor executable program instructions configured to cause the SCLC to perform operations when the processor executable program instructions are executed by the one or more SCLC processors, the operations comprising:

maintain the new metadata across each DCSI of the plurality of DCSIs; and

regenerate the out-of-data control signal to change the key slot in use from the currently used key slot to the different key slot to each next data connection segment interface.

14. The system of claim **13**, wherein the DSIHS non-transitory computer readable medium stores further processor executable program instructions configured to cause the DSIHS to perform further operations comprising: request a new key from a key management system; and update the different key slot with the new key.

15. The system of claim **13**, wherein the DCSI state machine non-transitory computer readable medium in each IHS E2EEE DCSI of the plurality of IHS E2EEE DCSIs stores further processor executable program instructions configured to cause the DCSI state machine to perform further operations comprising: maintain the state with the currently used and currently active metadata key slot, prior to the data source information handling system unlocking the encrypted data volume.

16. The system of claim **15**, wherein the DCSI state machine non-transitory computer readable medium of each DCSI of the plurality of IHS E2EEE DCSIs stores further processor executable program instructions configured to cause the DCSI state machine to perform further operations comprising: transition, the state with the currently used and active metadata key slot to use the different metadata key slot, in response to the out-of-data control signal to change the key slot in use from the currently used key slot to the different key slot; and maintain the currently active metadata key slot prior to the DSIHS unlocking the encrypted data volume.

17. The system of claim **13**, wherein the security chaining logic component SCLC in each IHS of the plurality of IHSs stores further processor executable program instructions configured to cause the SCLC to perform further operations comprising:

enable changing an encryption configuration state machine of each data DCSI to a state to use the different metadata key slot and make the different metadata key slot the active key slot, backward, from a last DCSI to a first DCSI, such that the last DCSI to the first DCSI progressively switch to the different key slot; and

enable encryption and decryption in each IHS E2EEE DCSI of the plurality of IHS E2EEE DCSIs forward, from a first DCSI to a last DCSI, such that encryption and decryption in each IHS E2EEE DCSI of the plurality of IHS E2EEE DCSIs use the different key slot.

18. The system of claim 13, wherein the rekeying is shallow, rekeying a key encryption key and any intermediate key.

19. The system of claim 13, wherein the rekeying is deep, rekeying a key encryption key, any intermediate key or keys and a data encryption key.

20. The system of claim 13, wherein the SCLC in each IHS of the plurality of IHSs stores further processor executable program instructions configured to cause the SCLC to perform further operations comprising:

in response to a first DCSI receiving a metadata update, interrupt and roll back an out-of-data control signal to change the key slot in use from the currently used key slot to the different key slot to a second DCSI; and

wait for a ready state in the metadata on the second DCSI to enable unlocking of the encrypted data volume.

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